Chapter | five

Memory Reconsolidation: Lingering Consolidation and the Dynamic Memory Trace

Cristina M. Alberini, Sarah A. Johnson, Xiaojing Ye
New York University, New York, New York

5.1 DEFINITIONS OF MEMORY CONSOLIDATION AND RECONSOLIDATION

5.1.1 Memory consolidation

Memory consolidation is a fundamental process of long-term memory formation, as, in fact, has been described to occur in a multitude of different types of memories, species, and memory systems. It refers to the stabilization process of a newly formed long-term memory. Initially, the memory is in a fragile state and can be disrupted by several types of interference, including behavioral, pharmacological, and electrical. Over time, the memory becomes resilient to these forms of interference through the process known as consolidation (Alberini, Bambah-Mukku, & Chen, 2012; Davis & Squire, 1984; McGaugh, 2000). Memory consolidation involves synaptic and broad cellular events that include transcriptional, translational, and post-translational mechanisms, as well as their feedback and feedforward regulation. These molecular changes start with the initial encoding and evolve with time because memory consolidation seems to occur in stages (Alberini, 2008, 2009; Dudai, 1996, 2012; McGaugh, 2000; Squire & Alvarez, 1995). For example, the consolidation of medial temporal lobe-dependent memories, in addition to the cellular consolidation events just described, also involves a redistribution of the memory trace, such that it transitions from hippocampal-dependent to hippocampal-independent (Squire & Alvarez, 1995; Squire, Clark, & Knowlton,

Memory Reconsolidation. http://dx.doi.org/10.1016/B978-0-12-386892-3.00005-6

2001; Squire, Stark, & Clark, 2004). Hence, memory consolidation appears to consist of multiple stages, which can be delineated based on the type of interference to which a memory trace is susceptible over time—for example, those that target molecular versus system mechanisms.

5.1.2 Memory reconsolidation

For several decades, it was believed that memory consolidation, at least the initial synaptic and cellular consolidation phase, was a unique process. A newly formed memory requires gene expression for several hours, after which it becomes stable and resilient to transcription and translation disruption (McGaugh, 2000). However, studies performed during the past decade, extending findings from the 1960s, demonstrate that the initial gene expression-dependent phase required for memory consolidation is not unique to a newly encoded memory. In fact, memories that have become resistant to disruption by inhibitors of gene expression, if reactivated, for example, by a retrieval event, again become temporarily labile. During this period, similar to what is seen during the initial consolidation, memories can be disrupted by molecular interference. Over time, the reactivated memories again become stable and resilient to disruption. This restabilization process has been termed *memory reconsolidation* (Alberini, 2005; Dudai & Eisenberg, 2004; Lewis, 1979; Nader, Schafe, & LeDoux, 2000; Sara, 2000). Although the term reconsolidation, as well as interpretations of findings about the reconsolidation process, is still highly debated, the existence of a temporal window of reacquired fragility following memory reactivation is supported by numerous studies. This post-reactivation fragility opens an opportunity for memory disruption or enhancement, and it has been reported in a wide range of species, from *Caenorhabditis elegans* to human, with respect to different types of learning, including aversive and appetitive, suggesting that it is a fundamental process underlying memory formation (Dudai & Eisenberg, 2004; Nader & Hardt, 2009).

5.2 CONSOLIDATION AND RECONSOLIDATION OF SINGLE TRIAL INHIBITORY AVOIDANCE CONDITIONING

5.2.1 What are the advantages of studying inhibitory avoidance and what information will we gain?

Inhibitory avoidance (IA) is a fear-based conditioning paradigm in which a single learning trial leads to the formation of a long-lasting memory. In this paradigm, animals learn to suppress a natural response, such as moving out of a brightly lit area into a darkened area (i.e., step-through inhibitory avoidance) or stepping down off a small starting platform (i.e., step-down passive avoidance) following conditioning to an aversive stimulus. A typical step-through IA task in rodents occurs as follows: During the training session, the animal is placed in a conditioning chamber that is divided into two compartments separated by an automated sliding door. One compartment is brightly lit, and the other is dark. The animal is placed in the lit chamber with the door closed. After a brief initial period to acclimate to the chamber (normally 10 sec), the door is opened, allowing the animal to cross to the dark compartment, upon which the door closes and a footshock is delivered via a grid floor. Memory retention can then be assessed at any time by returning the animal to the conditioning chamber and measuring its latency to enter the dark (shock) compartment. The retention level is a function of the intensity of the footshock used during training. In general, a maximum testing time is set (cut-off time; e.g., in our studies, 9 or 15 min), after which the animal is returned to its home cage. As a fear- and context-based conditioning paradigm, IA memory formation critically involves many brain regions similar to those mediating contextual fear conditioning, including the hippocampus, amygdala, anterior cingulate cortex, and prelimbic/infralimbic cortex (Ambrogi Lorenzini, Baldi, Bucherelli, Sacchetti, & Tassoni, 1997, 1999; Izquierdo *et al.*, 1997; Malin, Ibrahim, Tu, & McGaugh, 2007; McGaugh, 2004; McIntyre, Power, Roozendaal, & McGaugh, 2003; Zhang, Fukushima, & Kida, 2011). However, unlike contextual fear conditioning, IA is a form of instrumental learning (also called operant conditioning) because it requires that the subject takes a natural action during training and subsequently learns and decides to suppress the action (Staddon & Cerutti, 2003). IA provides several advantages for multilevel investigation of memory mechanisms, including molecular, behavioral, and translational. First, as described, it is easily learned with a single training trial, thus allowing the investigation of the temporal evolution of molecular and cellular events underlying the formation of a single memory trace. Second, this single trial induces a robust and long-lasting memory, thus allowing for tracing the reorganization of memory over time. Third, a single memory trace is also advantageous for isolating the responses associated with memory reactivation. Conversely, in tasks based on multiple training sessions, it is difficult, if not impossible, to determine the exact temporal evolution of the mechanisms underlying each learning trial and their contribution to the whole memory trace. IA has been widely accepted as a rodent model of human aversive/traumatic episodic memories. Hence, studying this task may provide important knowledge for designing and developing novel therapeutic approaches for disorders caused or precipitated by trauma and stress. At the same time, elucidating the mechanisms by which a temporal lobe-dependent memory is strengthened through consolidation may also offer the identification of strategies and tools that mediate memory enhancement and prevent memory loss. As such, it may lead to the discovery of new treatments that could combat numerous disease-based and aging-related cognitive impairments.

5.2.2 Mechanisms underlying the consolidation of IA memory

In order to understand the mechanisms and functions of IA memory reconsolidation, it is first important to highlight what is known with regard to the initial consolidation of IA memory. As with many other contextual and spatial (or, in humans, episodic and declarative) learning paradigms, both aversive and nonaversive, the consolidation of IA memory requires an intact hippocampus; in fact, lesion or inactivation of the hippocampus leads to both retrograde and anterograde amnesia (Lorenzini, Baldi, Bucherelli, Sacchetti, & Tassoni, 1996; Rossato *et al.*, 2004). Similar to that of many other memory paradigms investigated, the consolidation of IA requires post-translational, translational, and transcriptional mechanisms in both hippocampus and amygdala. In the hippocampus, IA consolidation requires the activation of cAMP-dependent protein kinase A (PKA) (Bernabeu, Cammarota, Izquierdo, & Medina, 1997), cAMP response element binding protein (CREB) (Bernabeu *et al.*, 1997; Impey *et al.*, 1998), the expression of CCAAT enhancer binding proteins (C/EBPβ and C/EBPδ isoforms) (Taubenfeld, Milekic, Monti, & Alberini, 2001; Taubenfeld, Wiig, Bear, & Alberini, 1999), mammalian target of rapamycin (mTOR) (Jobim *et al.*, 2012; Slipczuk *et al.*, 2009), and brain-derived neurotrophic factor (BDNF) (Slipczuk *et al.*, 2009), just to name a few. In addition, our laboratory found that in the hippocampus, IA consolidation also requires the activation of glucocorticoid receptors (GRs), whose engagement recruits the BDNF−CREB−C/EBPβ-dependent gene cascade (Chen, Bambah-Mukku, Pollonini, & Alberini, 2012). Moreover, hippocampal C/EBPβ-dependent target genes, whose regulation is critical for IA memory consolidation, include muscle-specific tyrosine kinase receptor (MuSK) (Garcia-Osta *et al.*, 2006) and insulin-like growth factor II (IGF-II) (Chen *et al.*, 2011). Similarly, in amygdala, IA consolidation requires *de novo* protein synthesis (Milekic, Pollonini, & Alberini, 2007), NMDA receptor- and AMPA receptor-dependent signaling (Bonini *et al.*, 2003; Jerusalinsky *et al.*, 1992), protein kinase C (Bonini, Cammarota, Kerr, Bevilaqua, & Izquierdo, 2005), CREB (Canal, Chang, & Gold, 2008), mTOR (Jobim *et al.*, 2012), corticotropin-releasing hormone (Roozendaal, Brunson, Holloway, McGaugh, & Baram, 2002), and C/EBPδ (Arguello, Ye, *et al.*, in press). However, unlike in the hippocampus, Induction of C/EBPβ is not required in the amygdala for IA consolidation (Tronel, Milekic, & Alberini, 2005; Milekic *et al.*, 2007) (Figure 5.1).

Our laboratory has been particularly interested in identifying the temporal evolution of these molecular cascades activated following IA learning in rats and their functional requirements in the hippocampus. We have found that the requirement of the CREB−C/EBP-target gene cascade, as well as *de novo* protein synthesis in general, lasts for more than 1 day after training, ends by 48 hr after training, and appears to involve feedforward autoregulation of BDNF and MuSK expression and function. Interfering with the initial phase of these changes completely prevents IA memory formation; interfering with the same molecular pathways later, but within the first 24 hr, leads to the establishment of a weaker memory that, however, rapidly decays. Finally, interfering with the same mechanisms at later times (e.g., 48 hr post-training) no longer has an effect on memory retention (Chen *et al.*, 2011; Garcia-Osta *et al.*, 2006; Taubenfeld *et al.*, 2001a, 2001b; Bambah-Mukku *et al.*, unpublished data).

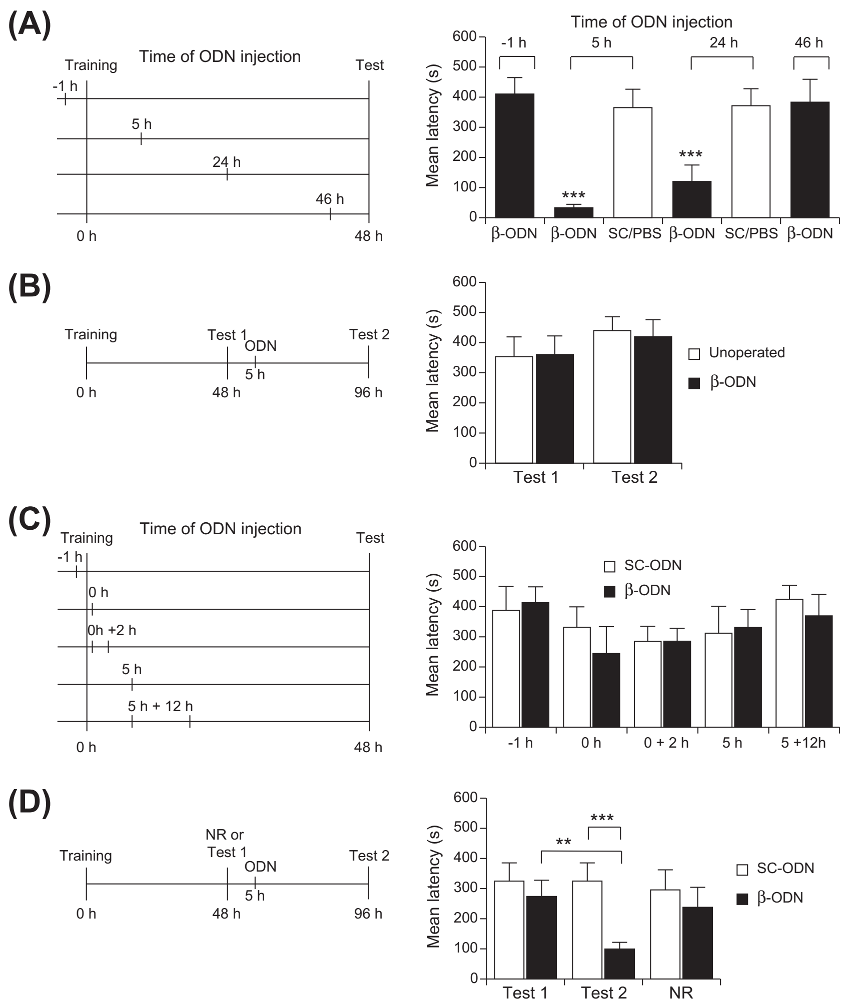


FIGURE 5.1 The requirement of C/EBPβ in dorsal hippocampus and basolateral amygdala (BLA) during consolidation and reconsolidation. Experimental schedules are shown beside each graph. (A) Dorsal hippocampal injection of C/EBPβ antisense oligodeoxynucleotide (β-ODN) at 5 or 24 hr after IA training significantly blocks memory retention at 48 hr, compared to scramble oligodeoxynucleotide (SC-ODN) or vehicle (PBS)-injected controls. Injection of β-ODN 1 hr before or 46 hr after training does not produce any significant memory impairment. (B) Dorsal hippocampal β-ODN injection at 5 hr after testing does not interfere with reconsolidation of IA memory. (C) C/EBPβ in the BLA is not required for IA memory consolidation. β-ODN or SC-ODN was injected bilaterally into the BLA at the following time points: 1 hr before training (−1 hr), immediately after (0 hr), 0 + 2 hr, 5 hr, or 5 + 12 hr after training. No significant difference in latency was observed among β-ODN- and SC-ODN-injected rats at any of the injection time points. (D) β-ODN injection into BLA 5 hr after reactivation impairs memory. Rats injected with β-ODN 5 hr after memory reactivation (Test 1) had significantly lower retention latencies when retested 48 hr later (Test 2, 96 hr post-training) compared to rats injected with SC-ODN or latencies at Test 1. $^{**}P < 0.01$, $^{***}P < 0.001$. *Source*: Reprinted by permission from Taubenfeld *et al.* (2001) (A and B) and from Milekic *et al.* (2007) (C and D).

Together, the molecular interference studies have been mainly restricted to investigating changes occurring rapidly after training and thus far, as mentioned previously, seem to suggest that these changes are completed by 24–48 hr after training. They have also been mostly carried out in hippocampus and amygdala but not yet in cortical regions, where, with the passage of time, memory consolidation seems to redistribute the memory trace.

Indeed, a number of studies noted that medial temporal lobe (MTL)-dependent memories undergo a consolidation process that can take weeks in animals and up to years in humans. Clinical studies of retrograde amnesia find that damage to the MTL structures, including hippocampus and adjacent cortices, impairs recent memories in a temporally graded manner but spares remote memories, whereas damage that involves the neocortex of lateral and anterior temporal lobes produces a constant loss of memory across the passage of time. Thus, the MTL seems to be necessary for the formation and maintenance of a memory for a limited time (McGaugh, 2000; Squire & Alvarez, 1995). This conclusion is further supported by findings from experimental studies in animal models (Squire *et al.*, 2001). For example, using contextual fear conditioning, Kim and Fanselow (1992) showed that lesion of the hippocampus on 1, 7, 14, or 28 days after training results in a temporally graded retrograde amnesia because 1-day memory is mostly impaired, whereas 28-day memory is spared. Similarly, in rats trained on a step-down IA task, blocking AMPA receptor signaling in the dorsal hippocampus and amygdala impaired a 1-day-old but not a 31-day-old memory (Izquierdo *et al.*, 1997; Quillfeldt *et al.*, 1996). Collectively, these findings suggest that there is a reorganization of the memory trace: Over time, the MTL regions are no longer essential for maintaining the information but, rather, memory storage and retrieval is mainly supported by the neocortex. This process is also termed system consolidation (Dudai, 2004).

In agreement with this temporally graded requirement of MTL in memory maintenance, neuroimaging studies in humans have shown that when subjects recall memories ranging from 1 to 30 years old, there is an anatomical redistribution of activity: The MTL structures exhibit greater activity during the recall of younger memories compared to older memories, whereas activity in the frontal, temporal, and parietal lobes shows the opposite pattern (Smith & Squire, 2009). Similarly, a series of studies has examined the activation of different brain regions by the retrieval of young versus old memories in rodents using functional mapping of neural activity (e.g., the induction of activity-related genes, uptake of 2-deoxyglucose, and neuronal firing patterns) with a variety of behavioral tasks (Frankland & Bontempi, 2005; Gusev & Gubin, 2010a, 2010b; Ross & Eichenbaum, 2006; Sacco & Sacchetti, 2010; Takehara-Nishiuchi, & McNaughton, 2008; Teixeira, Pomedli, Maei, Kee, & Frankland, 2006; Wiltgen, Brown, Talton, & Silva, 2004). In general, these studies find greater activation of dorsal hippocampus by the retrieval of recent compared to remote memories, whereas an opposite pattern is observed in a number of cortical regions.

Very few studies have explored the underlying mechanisms of this reorganization of memory traces over time. A recent report suggests that neurogenesis in adult hippocampus may regulate the temporal window within which contextual fear memory becomes independent from the hippocampus (Kitamura *et al.*, 2009). Other investigations found that dendritic spine growth in caudal anterior cingulate cortex correlates with and is required for remote contextual fear memory storage, whereas the hippocampus is required for driving spine growth in this region (Restivo, Vetere, Bontempi, & Ammassari-Teule, 2009; Vetere *et al.*, 2011).

Interestingly, there seems to be a critical link between sleep and system consolidation. *In vivo* recordings show that the patterns of neuronal firing in the hippocampus and other brain regions that occur during exploration of a novel environment or spatial tasks reoccur in the same order during subsequent sleep. This "implicit" memory reactivation mostly occurs during slow wave sleep. It has been suggested that during this stage of sleep, the neocortical slow oscillation may facilitate the interaction between the hippocampus and the neocortex, leading to redistribution of the memory trace (Born & Wilhelm, 2012; Diekelmann & Born, 2010; Marshall, Helgadottir, Molle, & Born, 2006; Molle, Yeshenko, Marshall, Sara, & Born, 2006; Sirota, Csicsvari, Buhl, & Buzsaki, 2003).

Animal studies also indicate that the rearrangement of the memory representation from hippocampal to cortical areas occurs together with a qualitative transformation. Using both social transmission of food preference and contextual fear conditioning tasks in rats, Winocur *et al.* (2007) report that when memories are assessed 1 day after training, performance is significantly better in the training context compared to a different context. However, as time elapses, this context specificity is lost and rats show increased memory expression in the different context. Similar results have also been reported for contextual fear conditioning in mice (Wiltgen & Silva, 2007). Thus, the initial memory, which would be highly detailed and context specific, seems to become one that is more general (Moscovitch *et al.*, 2005; Winocur, Moscovitch, & Bontempi, 2010).

An important question that remains to be addressed is whether the initial molecular consolidation that occurs in the MTL, such as that identified in the hippocampus of IA trained rats, redistributes with the trace as memory undergoes system consolidation, or whether different molecular/cellular mechanisms are engaged in the different brain regions and/or over time.

5.2.3 Mechanisms of IA memory reconsolidation

While the molecular mechanisms underlying memory consolidation were in the process of being uncovered, Nader *et al.* (2000) published their landmark study showing that memory for auditory fear conditioning could once again become sensitive to inhibition of *de novo* protein synthesis in the amygdala following retrieval, thus suggesting that memory can undergo a process of reconsolidation as well as identifying an underlying cellular process (see Chapter 2). This result changed the way the field viewed memory consolidation, which was, until then, mainly seen as a unitary stabilization process. This in turn prompted us to ask a number of questions about reconsolidation using the IA task in rats. First, are the initial IA consolidation and reconsolidation similar or different processes? Second, how does the passage of time influence the process of reconsolidation? Third, why does memory become labile after reactivation and need to reconsolidate? What function does reconsolidation serve? Finally, can we use reconsolidation mechanisms to disrupt pathogenic memories or to enhance healthy, adaptive memories?

Our studies, in agreement with studies examining different kinds of memories and different species, revealed that reconsolidation is a partial recapitulation of the initial consolidation (Alberini, 2005; Dudai & Eisenberg, 2004; von Hertzen & Giese, 2005). Specifically, in IA, *de novo* protein synthesis is required in both hippocampus and basolateral amygdala for IA memory consolidation, but it is only required in the basolateral amygdala for reconsolidation (Milekic *et al.*, 2007; Taubenfeld *et al.*, 2001a). In addition, C/EBPβ and C/EBPδ are required for both consolidation and reconsolidation of IA (although in different brain regions), and there are important distinctions in the critical functional implication of brain areas in the two processes. Specifically, C/EBPβ-dependent gene expression in the hippocampus is necessary for IA consolidation but not reconsolidation (Taubenfeld *et al.*, 2001a) (Figure 5.1). In contrast, C/EBPβ-dependent gene expression in the amygdala is required for IA memory reconsolidation but not for its initial consolidation (Milekic *et al.*, 2007; Tronel *et al.*, 2005) (Figure 5.1), whereas C/EBPδ is required in the hippocampus for IA consolidation and in the BLA for both consolidation and reconsolidation (Arguello, Ye, *et al.*, in press).

In agreement with the conclusions reached with these IA studies, other model systems, including other species, have found that consolidation and reconsolidation have distinct signatures. For example, Lee *et al.* (2004) reported that contextual fear conditioning consolidation requires hippocampal BDNF but not Zif268 expression, whereas the reconsolidation process requires hippocampal Zif268 but not BDNF expression (for more details, see Chapter 3). In addition, whereas protein synthesis is generally critical in the lateral amygdala for both consolidation and reconsolidation of auditory fear conditioning memory, inhibiting cap-dependent translation impairs consolidation but not reconsolidation of this memory (Hoeffer *et al.*, 2011). Finally, a comprehensive analysis of gene expression in the hippocampus of mice trained in contextual fear conditioning revealed that reconsolidation only induces a subset of molecules that are activated during consolidation (von Hertzen & Giese, 2005). Collectively, our studies in IA together with studies utilizing different memory tasks suggest that (1) consolidation and reconsolidation processes may engage distinct brain circuits and (2) even within the same brain region, they may recruit different or a subset of molecular mechanisms involved in the initial consolidation.

5.3 MEMORY RECONSOLIDATION AND THE PASSAGE OF TIME

Time is an extremely important parameter in memory formation and storage, allowing the selection of what is stored as long-term memory, regulating how long information will be stored, and allowing for continuous changes and updating of information. Memory consolidation takes time to be completed, evolves through different stages, and, thus, memory storage changes with time. How do these changes influence memory reconsolidation?

5.3.1 Temporal dynamics of memory reconsolidation

Understanding the temporal boundaries limiting reconsolidation is critical for understanding how memories are maintained and for the development of effective clinical approaches that target the reconsolidation process. Studies from our laboratory and others have revealed that the passage of time critically influences the ability of a memory to undergo reconsolidation; the older a memory becomes, the less susceptible it is to disruption following its reactivation. For example, in rat IA, 2- and 7-day-old memories are disrupted by protein synthesis inhibitors administered systemically after retrieval, whereas 14- and 28-day-old memories are resistant to the same treatment (Milekic & Alberini, 2002) (Figure 5.2). Similar findings have been reported for a variety of species, including rat (Bustos, Maldonado, & Molina, 2009), mouse (Boccia, Blake, Acosta, & Baratti, 2006; Frankland *et al.*, 2006; Suzuki *et al.*, 2004), chick (Litvin & Anokhin, 2000), and medaka fish (Eisenberg & Dudai, 2004); different types of behavioral paradigms, including both aversive and appetitive memories; and a range of pharmacological disruptions

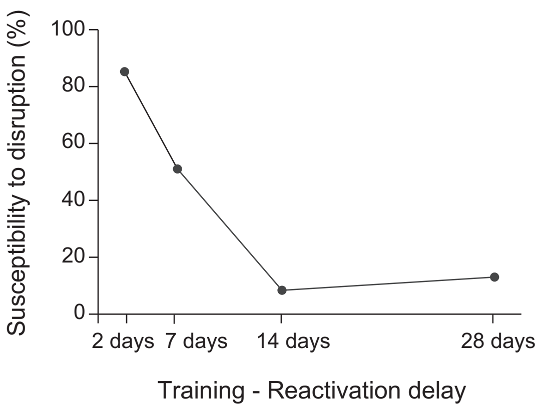


FIGURE 5.2 Temporally graded decreases of susceptibility to disruption of a reactivated memory. The percent (%) susceptibility to disruption by anisomycin injected immediately after retrieval of an IA memory was calculated using the following formula: [(mean latency [sec] of test one − mean latency [sec] of test two) / (mean latency [sec] of test one) × 100. *Source*: Reprinted by permission from Milekic *et al.* (2002).

(Alberini, 2011). Typically, in animal studies, the resilience to post-reactivation disruption occurs over a time window of weeks.

Importantly, as detailed later, memory reactivation and the post-reactivation fragility allow for not only memory disruption but also enhancement (Carbo Tano, Molina, Maldonado, & Pedreira, 2009; Chen *et al.*, 2011; de Souza *et al.*, 2004; Frenkel, Maldonado, & Delorenzi, 2005; Tronson, Wiseman, Olausson, & Taylor, 2006). In our lab, Chen *et al.* (2011) found that administration of IGF-II into dorsal hippocampus after retrieval of a 1-day-old memory, but not a 2-week-old memory, leads to a significant enhancement in memory retention, indicating that the enhancing effect of IGF-II in the hippocampus is restricted to a temporal window that exactly coincides with the reconsolidation-sensitive temporal period (Figure 5.3).

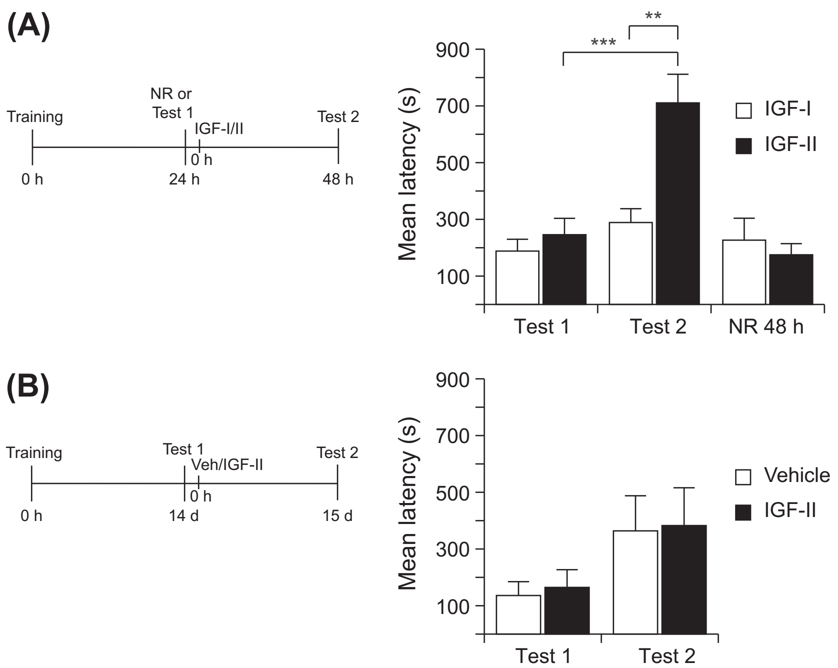
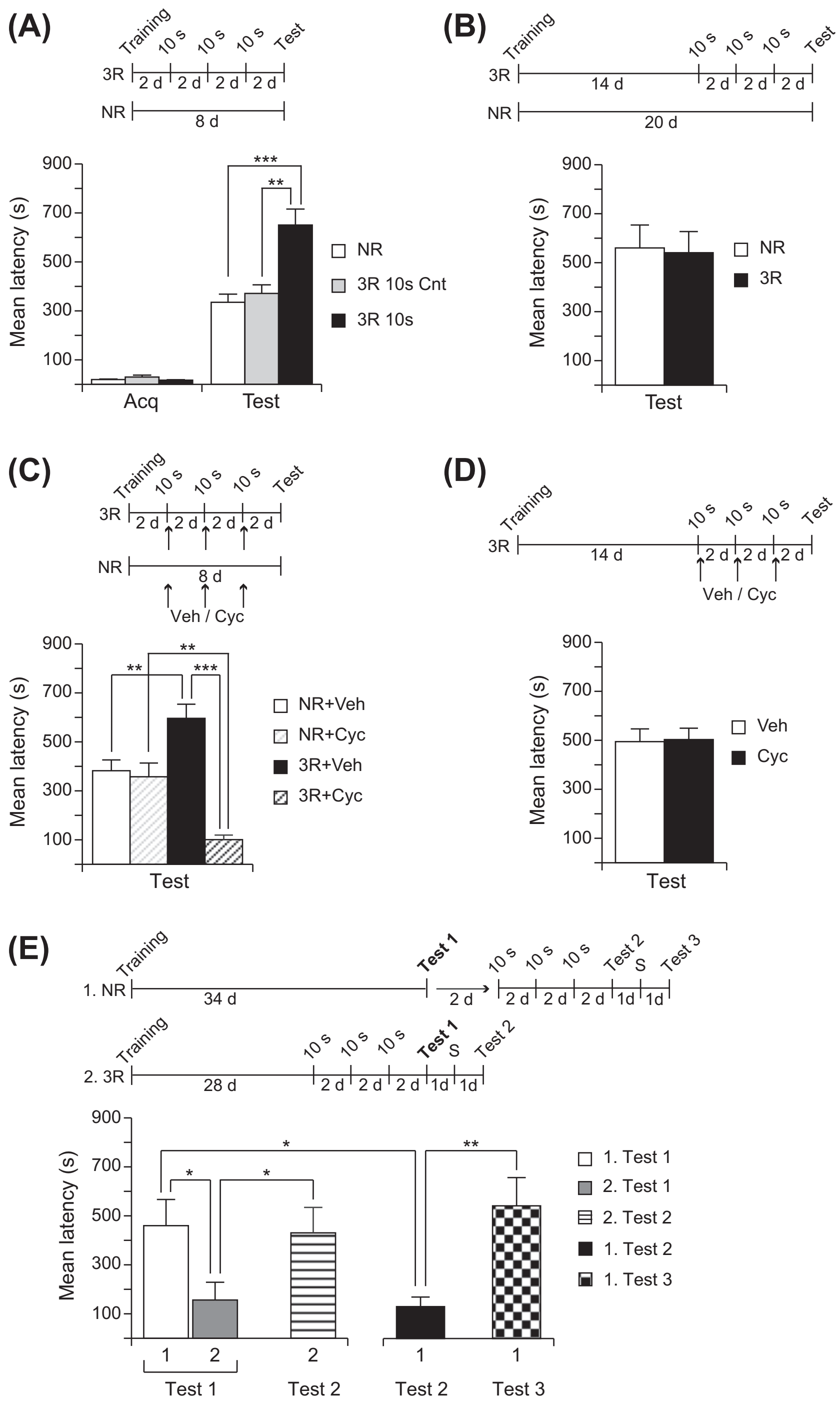


FIGURE 5.3 Post-retrieval IGF-II administration enhances memory, and the effect is temporally limited. Experimental schedules are shown beside each graph. (A) Rats were tested 24 hr post-IA training, and immediately after test (memory reactivation) they were injected with IGF-II or IGF-I bilaterally into dorsal hippocampus. Non-reactivated rats (NR) were trained and injected at the same time points without testing. Rats were tested again 48 hr post-training. Rats that received post-retrieval IGF-II injection had significantly enhanced retention latency compared to those injected with IGF-I or non-reactivated. $^{**}P < 0.01$, $^{***}P < 0.001$. (B) Rats trained in IA task were tested on 14 days post-training, and immediately after test they were injected with IGF-II or vehicle bilaterally into dorsal hippocampus. They were tested again 1 day later. IGF-II injection did not enhance the 14-day-old IA memory. *Source*: Reprinted by permission from Chen *et al.* (2011).

Other studies report that in some cases, even older memories can be vulnerable to disruption in certain circumstances and that post-retrieval memory fragility is a function of the strength of training, strength of reactivation, and passage of time (Suzuki *et al.*, 2004; for more details, see Chapter 6). Other investigations have also used long reactivation sessions to show that old memories or stronger memories can be disrupted by post-reactivation interference, at least to a certain extent (Bustos *et al.*, 2009; Diergaarde, Schoffelmeer, & De Vries, 2006; Eisenberg, Kobilo, Berman, & Dudai, 2003). However, all these results are in agreement with the general conclusion that the passage of time makes the memory stronger and less susceptible to post-reactivation interference. Furthermore, it is important to keep in mind that reactivation by retrieval of a conditioned response can (as in many studies of reconsolidation) evoke two processes—reconsolidation of the original memory trace and extinction learning, which teaches the animals that a conditioned stimulus (CS) is no longer paired with an unconditioned stimulus (US) and results in decreased expression of the conditioned response. Competition of the two traces seems to control the behavioral outcome (Eisenberg *et al.*, 2003). Long-lasting reactivation sessions preferentially evoke extinction over reconsolidation (Pedreira & Maldonado, 2003; Power, Berlau, McGaugh, & Steward, 2006; Suzuki *et al.*, 2004). Thus, when investigating whether and how the length of the reactivation session may influence the susceptibility of reactivated memories to disruption, it is important to exclude the possibility that memory disruption results from a facilitated extinction.

In line with the notion that memory evolves over time, using rat IA, Inda, Muravieva, and Alberini (2011) in our laboratory explored how the passage of time interacts with memory reactivation to regulate memory strength. It was found that multiple reactivations by brief exposure to the CS, which evoked memory reconsolidation, given within the first week after IA training led to significant memory strengthening. However, the same behavioral reactivations given 2 weeks after training, when memory already appeared to be stronger as a result of the passage of time, could not induce memory reconsolidation or strengthening. Interestingly, when the same reactivations were given 4 weeks after training, when memory had partially decayed, they actually produced extinction instead of reconsolidation and strengthening (Figure 5.4). These findings not only support the hypothesis that the function of memory reconsolidation is, at least in part, to strengthen memory retention during the temporal window of system consolidation (discussed later) but also show that memory storage is temporally dynamic. First, over time, without explicit reactivations, memory strengthens and becomes refractory to both reactivation-dependent interference and strengthening. Second, reactivation of a young memory preferentially triggers reconsolidation over extinction, whereas reactivation of an old memory preferentially triggers extinction over reconsolidation. These results have important clinical implications because the same post-reactivation intervention may affect different processes, depending on the age and state of the memory.


(A)
Training
10 s
10 s
10 s
Test
3R
2 d 2 d 2 d 2 d
NR
8 d
Mean latency (s)
NR
3R 10s Cnt
3R 10s
Acq
Test
(B)
Training
10 s 10 s 10 s Test
3R
14 d
2 d 2 d 2 d
NR
20 d
NR
3R
Test
(C)
Veh / Cyc
NR+Veh
NR+Cyc
3R+Veh
3R+Cyc
(D)
Veh
Cyc
(E)
1. NR
34 d
Test 1
2. 3R
28 d
Test 2
Test 3
S
1. Test 1
2. Test 1
2. Test 2
1. Test 2
1. Test 3

Some authors disagree with this conclusion. They view reconsolidation as a general process that occurs following memory retrieval without temporal constraints, and they explain temporal boundaries as experimental limitations (Nader & Einarsson, 2010). For example, Nader *et al.* (2000) showed that both 2-day- and 2-week-old auditory fear conditioning memories in rats were disrupted by post-retrieval bilateral injection of the protein synthesis inhibitor anisomycin into the lateral and basal nuclei of the amygdala. Similarly, injection of Zif268 antisense before re-exposure to a cocaine-associated cue at 3 and 27 days after cocaine self-administration impaired subsequent cocaine-seeking behaviors (Lee, Milton, & Everitt, 2006). Furthermore, in contrast to some studies showing that contextual fear memories in mice become less sensitive to post-retrieval interference as memories age (Frankland *et al.*, 2006; Suzuki *et al.*, 2004), Debiec *et al.* (2002) reported that 3- to 45-day-old contextual fear memories in rats were disrupted by post-retrieval bilateral injection of anisomycin into hippocampus. Interestingly, lesion of the hippocampus 45 days after training did not result in memory impairment, suggesting that by this time the contextual fear memory is independent of the hippocampus; however, reactivation of the memory caused the trace to again become dependent on the hippocampus. It is not clear what contributes to these conflicting results. One

FIGURE 5.4 Memory reactivation leads to reconsolidation and strengthening in a recent memory and extinction in a remote memory. Experimental schedules are shown beside each graph. (A) Rats were trained (Acq), and 2 days later memory was reactivated with three 10—sec context exposures with a 2-day inter-reactivation interval (3R). A control group was trained and received three 10-sec exposures to a different context (Context B; Cnt), whereas another control group remained in the home cage (NR). At testing (Test), the 3R group showed a significant increase in latency compared with the NR group. (B) Rats that underwent three reactivations (3R) 14 days after training or NR had similar, strong latencies when tested 20 days after training. (C) Rats were trained and received 3R beginning 2 days later, or NR, as in panel A. Both groups were injected immediately after each reactivation, or at a paired time point, with either cycloheximide (Cyc) or vehicle (Veh), and they were tested 8 days after training. At testing, 3R significantly increased latency, whereas Cyc prevented this increase. (D) Rats underwent the same training and reactivation protocol as in panel B, and they were injected with either Cyc or Veh after each reactivation as in panel C. At testing, no difference was found between groups. (E) Reactivation of a 4-week-old memory led to a facilitation of extinction. Rats were trained and divided into two groups. The first group (1) was tested 34 days after training. The second group (2) underwent 3R starting 28 days after training. Both groups were tested 34 days after training (test 1). At test 1, group 2 had a significantly lower latency than group 1. To verify whether this loss of memory was due to extinction, group 2 received a footshock reminder (S) 1 day after test 1 and was tested again the following day (test 2). Group 2 showed reinstatement of memory retention as, in fact, at test 2 showed significantly higher retention than at test 1. After test 1, group 1 received 3R and was tested 2 days later (test 2). Again, these animals showed extinction because their latency was significantly lower than that at test 1. A reminder footshock given 1 day later resulted in significant reinstatement of the memory at test 3 the following day. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$. *Source*: Reprinted with permission from Inda *et al.* (2011).

possibility, as discussed previously, is that memory reactivation triggers both reconsolidation and extinction processes. At least under certain conditions, inhibiting hippocampal protein synthesis or other forms of molecular interference can result in enhanced extinction of contextual fear (Cai, Blundell, Han, Greene, & Powell, 2006; Fischer, Sananbenesi, Schrick, Spiess, & Radulovic, 2004). Thus, it is important to determine whether the retrieval-dependent impairment in memory expression is due to blocking reconsolidation or facilitating extinction. In addition, the temporal window for the stabilization process and circuitry storage distribution is likely to be different for different memories and modalities of training (e.g., training intensity). However, the answer may be that both conclusions are correct and that different types of memories follow different consolidation processes or stages. For example, one possibility that remains to be investigated is that the temporal window of reconsolidation in medial temporal lobe-dependent memories is limited because these memories undergo system consolidation and trace redistribution, whereas cue-dependent memories (e.g., implicit memories) do not undergo such redistribution, and their storage might therefore be more restricted and mechanistically distinct. A different distribution of the memory storage, together with a change in quality of the mechanisms underlying memory storage, may be the answer to these controversies.

5.4 FUNCTIONS OF MEMORY RECONSOLIDATION: UPDATING AND STRENGTHENING

Why do memories undergo reconsolidation? Given that during reconsolidation memory may be lost if interference occurs, what purpose does the post-reactivation destabilization of a memory trace serve? Two non-mutually exclusive hypotheses have been proposed to address this question: (1) Reconsolidation serves as an opportunity for memory updating, allowing new information to be incorporated into existing memory traces, and (2) reconsolidation provides an opportunity or a mechanism for strengthening and enhancing memory retention.

5.4.1 Reconsolidation and memory updating

At least two forms of memory updating have been investigated to date: the linking of novel, distinct information to an old memory trace and the addition of information regarding the same experience (i.e., incremental learning). Although both can be defined as memory updating, these processes are likely controlled by very different underlying mechanisms.

With respect to the first form of memory updating, that of incorporation of new information into an existing trace, in our laboratory, Tronel *et al.* (2005) found that second-order conditioning—that is, the association of a novel and distinct conditioned stimulus (CS2) to a previously formed association (CS1−US)—is mediated by consolidation mechanisms. This conclusion was reached using a molecular requirement that doubly dissociates consolidation from reconsolidation of IA memory in rats, namely that C/EBPβ is required in the dorsal hippocampus for consolidation but not reconsolidation, whereas C/EBPβ is required in the basolateral amygdala for reconsolidation but not consolidation.

Specifically, inhibition of C/EBPβ in the amygdala after reactivation of IA memory (CS1−US) in a new context (i.e., CS2) did not prevent the formation of the CS2−(CS1−US) association through second-order conditioning because rats subsequently tested in this new context showed clear memory. However, as would be predicted, inhibition of C/EBPβ in the amygdala interfered with reconsolidation of the first trace (i.e., CS1−US) because rats no longer showed memory for the original context (Figure 5.5). In contrast, second-order

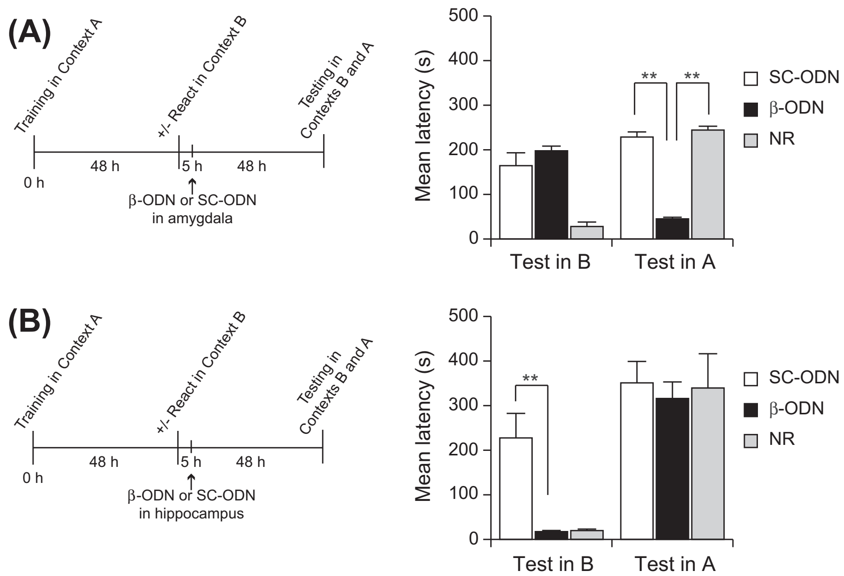


FIGURE 5.5 Consolidation, but not reconsolidation, mechanisms are required to associate new information with a reactivated memory. Experimental schedules are shown beside each graph. Rats were trained in IA in a shuttlebox (context A), which included a contextual cue (light turned on). Memory was reactivated with the same contextual cue (light turned on) 48 hr later in an otherwise distinct shuttlebox context (context B). Memory for either the original trace or the second-order trace was assessed 48 hr after reactivation by testing the latency in context A or context B, respectively. (A) Amygdala injection of C/EBPβ antisense oligodeoxynucleotide (β-ODN) after memory reactivation disrupted the memory for context A without affecting the memory for context B. In contrast, the same treatment strongly impaired memory of context A, compared to SC-ODN injection. Memory of context A was not affected in rats that did not receive reactivation. (B) Hippocampal injection of β-ODN blocked the formation of an association between new and reactivated information without affecting the stability of the reactivated memory. β-ODN injection into the hippocampi of rats 5 hr after memory reactivation significantly impaired memory retention for context B 48 hr later, compared with SC-ODN injection. However, the same β-ODN injection did not affect the memory of context A, which remained similar to that of both control groups that received either SC-ODN injection after reactivation or β-ODN injection in the absence of reactivation. **$P < 0.01$. *Source*: Reprinted with permission from Tronel *et al.* (2005).

conditioning was prevented by inhibition of C/EBPβ in the dorsal hippocampus, suggesting that this memory underwent a new process of consolidation. Disruption of C/EBPβ in the hippocampus did not affect the reconsolidation of the original memory (Figure 5.5). Together, these data indicate that reconsolidation does not mediate memory updating through second-order conditioning but, rather, that updating engages a new consolidation process. In fact, disrupting the reconsolidation of the original CS1—US memory does not affect the new CS2—(CS1—US) association, supporting the conclusion that the two memories exist completely independent of each other.

Our findings that memory updating requires consolidation, for the purpose of adding new information to an existing trace, suggest that this form of updating is not a primary function of reconsolidation. In fact, given that memories are continuously updated throughout the life span, how could a reconsolidation-based mechanism that is temporally restricted and mainly occurs when a memory is recent be the process that mediates all updating of old memories? New information is constantly integrated into a network of memory traces, and this seems to occur via new consolidation processes. We speculate that for very long-lasting memories, it would be less adaptive that the updating of an existing memory would transform the original memory into a new one, whereas it seems more advantageous that to provide behavioral adaptability and choices, the old memories continue to coexist with new memories that comprise both old and new linked information. An example of this situation is extinction, whereby a new extinction memory trace coexists with the original conditioning trace. Memory updating, consolidation, and reconsolidation are all important pieces of the dynamic memory trace and storage, and understanding these processes and how they evolve over time is key for developing effective treatments of psychopathologies, whether they are pharmacological, behavioral, or the combination of approaches. This knowledge will also be critical for developing more effective psychotherapeutic treatments (see Chapter 14).

Data from other memory paradigms and other model organisms support the conclusion that integrating new information occurs through consolidation and that two related traces can continue to exist in parallel after updating. For example, Debiec *et al.* (2006), using rat auditory fear conditioning, found that disrupting the reconsolidation of a CS1—US memory by inhibiting amygdala protein synthesis after reactivation does not affect the formation of a new, related second-order association (i.e., CS2—CS1—US). Similar results were also found with associative memories in the crab *Chasmagnathus* (Suárez, Smal, & Delorenzi, 2010), as well as in humans (Forcato, Rodríguez, Pedreira, & Maldonado, 2010). Hence, we conclude that the primary function of reconsolidation must be different from that of linking an existing, reactivated memory to a novel, distinct experience. In general terms, we can infer that memory updating via formation of complex networks requires memory reactivation but not reconsolidation, and that reconsolidation of a reactivated memory does not alter the entire network of updated associative memories.

Other studies have tested whether reconsolidation mediates memory updating by examining the contribution of the repetition of similar training trials compared to that of encoding new information. These studies concluded that with multiple learning trials or reactivations, memory does indeed become labile. However, the fragility is seen only when learning is in a non-asymptotic mode. Specifically, inhibition of protein synthesis after repeated training trials during a non-asymptotic phase of learning returns memory performance to a pretraining chance level, suggesting that the original trace had remained labile. This effect has been shown using spatial learning (Meiri & Rosenblum, 1998; Touzani, Puthanveettil, & Kandel, 2007) and motor learning (Luft, Buitrago, Kaelin-Lang, Dichgans, & Schulz, 2004; Luft, Buitrago, Ringer, Dichgans, & Schulz, 2004). In contrast, when performance on a task has reached ceiling levels, and memory has therefore reached an asymptotic phase, memory is resistant to disruption by post-reactivation amnesic treatments (Morris *et al.*, 2006; Rodriguez-Ortiz, De la Cruz, Gutiérrez, & Bermudez-Rattoni, 2005; Winters, Tucci, & DaCosta-Furtado, 2009).

Interestingly, memory becomes labile once again if conditions are adjusted to induce a new phase of encoding. For example, using a delayed matching-to-place task in rats requiring that they locate a platform in a water maze, Morris *et al.* (2006) found that reconsolidation of a well-trained spatial memory is engaged only when there is a shift in location of the escape platform, representing a mismatch in contextual information that triggers updating of their cognitive representation of space. With a combination of taste recognition and taste aversion learning in rats, Rodriguez-Ortiz *et al.* (2005) found that inhibition of protein synthesis in the gustatory cortex disrupted a well-trained memory for saccharin taste only when malaise was introduced in conjunction with reactivation, another situation in which a mismatch between prior and current conditions triggers updating of the memory for familiar taste. Winters *et al.* (2009) noted a similar effect using spontaneous object recognition in rats: Inhibition of NMDA receptor function interfered with expression of the original object memory when reactivation introduced new contextual information but not when reactivation was identical to the initial training sessions.

As in studies discussed previously in relation to memory updating, it is unclear whether disruption of the memory occurs, because during reactivation the rat associates old information (e.g., the water maze, saccharin taste, or objects) with novel information (e.g., a new platform location, malaise, or new contextual cues), and the interference disrupts the consolidation of a new memory trace containing information about the contextual mismatch rather than the reconsolidation of the original trace. Much like the case for extinction learning in conditioning paradigms, two memory traces could potentially coexist and both contribute to the expression of behavior in the tasks described previously. Also, because both consolidation and reconsolidation are sensitive to similar forms of molecular disruption, using an interference approach that does not doubly dissociate the two processes cannot irrefutably demonstrate that reconsolidation mediates this type of updating.

Thus, an alternative interpretation from that offered by authors who suggest that reconsolidation mediates memory updating when novel information is linked to an existing trace is twofold. First, in agreement with the authors' conclusions, during a learning curve, post-trial application of amnesic treatments disrupts memory retention only when the memory is not at an asymptotic level. However, when retention has reached an asymptotic level and no further learning or increased retention is evident, a previously consolidated memory remains stable and resistant to disruption. Second, if new events are then presented and associated with this memory, a new trace is formed that is labile because it undergoes consolidation. If part of the old information is incorporated in a new trace, its retention might be disrupted if the new trace is challenged by amnesic treatments, following the rules of the predominant active trace (Eisenberg *et al.*, 2003).

Hence, we speculate that for memories that undergo system consolidation through the medial temporal lobe, reconsolidation is engaged only to change the strength of the original memory; in other words, reactivation of the memory strengthens or further consolidates the original memory without changing its content.

5.4.2 Reconsolidation and memory strengthening

Although inducing fragility of a memory trace through reactivation or retrieval can open it to interference, it appears to offer an equivalent opportunity for enhancement and strengthening. In fact, we propose that during the temporal window in which system consolidation occurs, memory strengthening through reactivation or retrieval is the chief function of reconsolidation.

Although it is intuitive to predict that repeated training sessions would reinforce memory formation and result in memory strengthening, as observed in situations of asymptotic memory performance described previously, it is less obvious to expect the same effect from non-reinforced exposures. As discussed in previous sections, presentation of stimuli under non-reinforced conditions (i.e., in the absence of the unconditioned aversive stimulus or reinforcer, such as a footshock or food reward; US) leads to extinction, a distinct learning process that is also protein synthesis dependent (Berman & Dudai, 2001; Power *et al.*, 2006; Santini, Ge, Ren, Peña de Ortiz, & Quirk, 2004). However, brief exposure to a non-reinforced reminder may not result in extinction but, rather, may preferentially induce reconsolidation and enhancement of the memory. In other words, reactivation of a memory can result in its strengthening through reconsolidation or suppression through extinction, and the bias toward either of these processes would be determined by the dominant state of the trace (Eisenberg *et al.*, 2003).

As mentioned previously, we have found that brief, repeated 10-sec exposures to the IA context (i.e., memory reactivations) given during the first week after training, while the IA memory trace is still undergoing system consolidation, lead to memory strengthening and prevent forgetting (Inda *et al.*, 2011). This strengthening of IA memory is contingent upon *de novo* protein synthesis because memory retention is disrupted by systemic injections of the protein synthesis inhibitor cycloheximide prior to each of the three reactivations. However, when the same brief reactivations were given 2 weeks after IA training, at a time when the memory is stronger and known to be resistant to the effects of protein synthesis inhibitors administered after retrieval (Milekic & Alberini, 2002), we found no effects of reactivation or post-retrieval molecular interference (Figure 5.4). Interestingly, with older IA memories beginning 28 days after training, a time at which memory has decayed, the three reactivations led to extinction (Inda *et al.*, 2011) (Figure 5.4). Thus, whereas reactivation of a recent memory leads to its strengthening through reconsolidation, the reactivation of a remote memory promotes its extinction, suggesting there is in fact a temporal boundary on the time period within which reconsolidation can be harnessed to improve memory. This period may overlap with the period of system consolidation.

The fact that memory strengthens over time without explicit reactivations leads us to speculate, as others have (Diekelmann, Büchel, Born, & Rasch, 2011), that reconsolidation is induced through implicit reactivations, and that this may be the mechanism through which memory becomes stronger and resilient to disruption. In support, studies of the incubation of conditioned fear, anxiety, and conditioned reward-seeking suggest that memory is strengthened over time, in the absence of explicit retraining, reactivation, or retrieval (Bindra & Cameron, 1953; Grimm, Fyall, & Osincup, 2005; Lu, Grimm, Hope, & Shaham, 2004; Pickens *et al.*, 2011; Pickens, Golden, Adams-Deutsch, Nair, & Shaham, 2009). Although the mechanisms underlying regulated implicit reactivation are far from clear, sleep appears to play an important role in the consolidation of long-term memories and may provide an opportunity for lingering consolidation to take place over time (Diekelmann & Born, 2010; Graves, Pack, & Abel, 2001; Stickgold, Hobson, Fosse, & Fosse, 2001).

Overall, it is clear from our work and the work of others that reconsolidation serves the important function of strengthening memory traces when partially or fully reactivated, but only within a defined window of time after the initial training. Given these temporal boundaries, interesting questions arise regarding the interplay of reconsolidation and system consolidation, and the temporal boundaries of system consolidation.

5.5 MEMORY STRENGTHENING VIA RECONSOLIDATION: MECHANISMS AND POTENTIAL APPLICATIONS

5.5.1 Molecular, cellular, and systems mechanisms of strengthening through reconsolidation

Very little is known about the molecular and cellular mechanisms responsible for memory enhancement through reconsolidation. However, targeting these mechanisms may represent an important strategy for delaying memory loss and combating cognitive impairments. Early studies suggested that one potential mechanism through which reconsolidation may strengthen memory is by increasing attention and arousal. DeVietti, Conger, and Kirkpatrick (1977) demonstrated that stimulation of the mesencephalic reticular formation, targeting noradrenergic fibers originating in the locus coeruleus, in conjunction with memory retrieval, enhanced subsequent memory performance. Others went on to demonstrate that direct stimulation of the locus coeruleus during retrieval enhanced memory for a maze task and that this enhancement was mediated by β-adrenergic receptor activation (Devauges & Sara, 1991; Sara, 2000). Indeed, it is possible that memory reactivation re-engages neuromodulatory processes affecting components of the trace, which in turn facilitates retrieval and memory strengthening.

More recent molecular investigations have identified factors that, if given in tandem with memory reactivation, enhance memory retention, suggesting that they boost reconsolidation mechanisms. For example, activation of PKA in the amygdala by infusing 6-BNZ−cAMP after retrieval of an auditory fear conditioning memory leads to subsequent enhancement of the memory (Tronson, Wiseman, Olausson, & Taylor, 2006). Similarly, increasing brain levels of angiotensin II following reactivation of a contextual fear memory in the crab *Chasmagnathus* leads to strengthening of the trace (Frenkel, Maldonado, & Delorenzi, 2005). It is interesting to note that in this particular study, the authors induced release of angiotensin II through water deprivation, an ethologically relevant circumstance during which identifying and remembering potential survival tactics would be beneficial. This finding suggests that factors released in response to internal states at times when memory activation is critical may support reconsolidation, and particularly memory strengthening (or, in other conditions, perhaps extinction).

Furthermore, as mentioned previously, we have recently shown that infusion of IGF-II in the dorsal hippocampus after retrieval of IA memory significantly strengthens the memory, and the effect occurs only at the times at which reconsolidation takes place (i.e., 1 but not 14 days after training; Chen *et al.*, 2011) (Figure 5.3).

The identification of mechanisms underlying memory strengthening via reconsolidation is important and has immediate translational potential. For example, normal aging is accompanied by cognitive decline, and its effects can significantly impede day-to-day functioning in older adults. Interestingly, age-related memory loss is not ubiquitous, in that it does not affect all individuals, and does not impact all types of memory. Rather, deficits most often manifest in declarative or episodic memory function—that is, explicit memory for information encountered during the course of daily living. These deficits seem to affect most profoundly the processes of long-term memory. In fact, when tracked over the human life span, short-term and early phases of long-term memory are spared, whereas the formation of long-term memory is significantly attenuated (Cansino, 2009). A similar pattern is observed in studies of middle-aged and older animals, such that working and short-term memory in hippocampal-dependent tasks remain intact, whereas probe tests given after longer retention intervals indicate memory loss (Bizon *et al.*, 2009; Countryman & Gold, 2007; Gallagher & Rapp, 1997; LaSarge *et al.*, 2007; Robitsek, Fortin, Koh, Gallagher, & Eichenbaum, 2008).

This pattern of memory decline suggests that consolidation and reconsolidation gradually become less effective as we age. Understanding the impact of normal aging on mechanisms of memory consolidation, and particularly reconsolidation, is therefore important because reconsolidation mechanisms could potentially be boosted through memory retrieval or pharmacological interventions to lessen the impact of age-related memory loss. Although studies investigating the effects of normal aging on molecular mechanisms of consolidation and reconsolidation are sparse, several lines of evidence suggest that decline of these functions could be linked to impairment of long-term memory on hippocampal-dependent tasks. For example, loss of input from the entorhinal cortex to the dentate gyrus via fibers of the perforant path has been observed in older rats, and this decreased connectivity is reflected by a failure to induce and maintain long-term potentiation, relative to younger rats (Burke & Barnes, 2006). Induction of factors required for memory consolidation in the hippocampus is also compromised in some circumstances; aged rats do not show increased hippocampal expression of pCREB and C/EBPβ in response to contextual fear conditioning (Monti, Berteotti, & Contestabile, 2005), and aged rats impaired in a spatial Morris water maze task show reduced CREB protein levels, relative to young and unimpaired rats (Brightwell, Gallagher, & Colombo, 2004).

5.6 PHARMACOLOGICAL DISRUPTION OF RECONSOLIDATION: WEAKENING PATHOGENIC MEMORIES

The re-emergence of research focusing on memory reconsolidation has stimulated interest in the medical community, particularly among mental health professionals treating disorders that are based on maladaptive memories. Two examples that have been investigated in animal models and in humans are post-traumatic stress disorder (PTSD) and drug addiction, both of which can be viewed as the result of dysfunctional memory processes.

PTSD can develop after an exposure to an emotionally or physically traumatic event, and the illness is characterized by repeated, intrusive memories of the experienced trauma. Patients suffering from PTSD have difficulty sleeping and feel detached or estranged, and these symptoms can be so severe and persistent so as to significantly impair the person's daily life. A behavioral model of PTSD onset has been proposed: The traumatic event (US) triggers a strong hormonal stress response, which mediates the formation of a robust and enduring memory for the trauma. Subsequent recall of the event in response to cues and reminders (CS) releases more stress hormones (conditioned response) and even further consolidates the memory leading to PTSD symptoms, such as flashbacks, nightmares, and anxiety (Pitman & Delahanty, 2005). The persistence of PTSD can be explained in terms of a trauma-induced strengthening of the memory trace. Specifically, it is hypothesized that noradrenergic hyperactivity and stress hormones facilitate the encoding and consolidation of the memory (O'Donnell, Hegadoren, & Coupland, 2004; Pitman, 1989; Yehuda *et al*., 2006).

Although it is not possible to precisely reproduce PTSD in an animal model, fear conditioning in rodents can be used to mimic and elucidate some aspects of PTSD, including the processing of fearful stimuli and the encoding of emotional memory. Thus, it is possible to closely approximate some of the PTSD symptoms in an animal model that can be used for preclinical research. In principle, forms of pharmacological and behavioral interference thus far found to be effective in disrupting fear memory reconsolidation or enhancing extinction (not discussed in this chapter) could potentially be useful for identifying new treatments that can be tested in clinical trials. Some of the compounds found to disrupt reconsolidation, including antagonists of β-adrenergic receptor, glucocorticoid receptors (GRs), or rapamycin, are already used in clinical pharmacology for treating other diseases. Hence, they are the most readily available potential therapies for targeting reconsolidation in PTSD and addiction.

Antagonists of the β-adrenergic receptor, such as propranolol and rapamycin, have already been explored at clinical levels. Propranolol, which is most commonly used to treat hypertension, has been administered in concert with the retrieval of a fearful or traumatic event in both animals and humans. Przybyslawski, Roullet, and Sara (1999) provided one of the first reports on the effect of propranolol on memory reconsolidation. Using rat IA, these authors found that systemic administration of propranolol after the reactivation of an IA memory disrupted the memory on subsequent tests. However, we recently reached the opposite conclusion. Because the shock used by Przybyslawski *et al.* was very weak (0.2 mA), we set out to determine whether the reconsolidation of a memory induced by a greater shock, which may more closely approximate a traumatic event, was sensitive to propranolol treatment. Hence, we tested the effect of the same propranolol treatment in rats given either before or after the retrieval of an inhibitory avoidance memory evoked by a 0.6- or 0.9-mA footshock. We found that although these memories could be disrupted by several other treatments, such as anisomycin or the GR antagonist RU38486, propranolol had no effect (Muravieva & Alberini, 2010). These divergent results could be related to several differences in the protocols used. However, to provide solid preclinical information, it is important that the potency and generalization of the propranolol treatment on fear memories are established. Using auditory fear conditioning, Debiec and LeDoux (2004) found that propranolol injected either systemically or into the lateral nucleus of the amygdala after reactivation of a 1-day- or 2-month-old memory weakened the fear response when tested 48 hr later (see also Chapter 4). In agreement, in recent studies based on contextual—auditory Pavlovian fear conditioning, we found that systemic propranolol injection following a retrieval elicited by cue exposure interferes with the reconsolidation of both cued and contextual fear conditioning. On the other hand, propranolol administered after contextual reactivation only affects contextual fear conditioning and has no effect on the auditory fear conditioning. Thus, it seems that the efficacy of systemically administered propranolol in disrupting the reconsolidation of fear memories might be limited (Muravieva & Alberini, 2010).

Other studies have reported divergent effects of propranolol on memory reconsolidation (for discussion of human studies, see Chapter 9). Whereas the reconsolidation of eyeblink conditioning potentiation and conditioning to natural or drug-associated reward is disrupted by propranolol (Diergaarde *et al.*, 2006; Kindt, Soeter, & Vervliet, 2009; Milton, Lee, & Everitt, 2008), the reconsolidation of appetitive Pavlovian memories in rats (Lee & Everitt, 2008) and that of neutral and emotional verbal memories in humans (Tollenaar, Elzinga, Spinhoven, & Everaerd, 2009) are not. Interestingly, declarative measures for the acquired contingency between the CS and the US (Kindt *et al.*, 2009) are insensitive to propranolol treatment, but the fear response is sensitive. Hence, as Kindt *et al.* suggested, propranolol may target the fear response but not the cognitive or explicit components of that response. Further studies should address this question and will be important to determine whether blocking the noradrenergic mechanisms during the reconsolidation of fear/traumatic memories indeed mainly disrupts the emotional component without much interfering with the explicit/declarative representation.

Another pharmacologically targeted pathway for the potential treatment of PTSD, which we have investigated using IA, is the glucocorticoid pathway. The endogenous stress hormone corticosterone bidirectionally modulates memory retention (McGaugh & Roozendaal, 2002; Roozendaal *et al.*, 2002). Low doses increase memory retention, whereas high doses disrupt it. Recent studies have reported that glucocorticoids, when administered after the reactivation of a contextual fear memory, have an amnesic effect on the original memory and provide evidence that a possible mechanism for this effect is an enhancement of extinction of the expression of the original memory (Abrari, Rashidy-Pour, Semnanian, & Fathollahi, 2008; Cai, Blundell, Han, Greene, & Powell, 2006). A study from our laboratory, however, revealed that intra-amygdala blockade of glucocorticoid receptors with the antagonist mifepristone (RU38486) after reactivation of IA memory in rats significantly disrupts the original memory, probably interfering with its reconsolidation process (Tronel & Alberini, 2007). Further investigations have also shown that systemic administration of RU38486, either before or after retrieval, consistently weakens IA retention in a dose-dependent manner (Nikzad, Vafaei, Rashidy-Pour, & Haghighi, 2011; Taubenfeld, Riceberg, New, & Alberini, 2009). The efficacy of treatment appears to be a function of the intensity of the initial trauma. Highly traumatic IA memories are not disrupted with RU38486 administered systemically after retrieval given 1 day after training. However, these memories become sensitive to this intervention after 1 week. Hence, according to the results in the IA paradigm, for stronger or highly traumatic memories, there is a temporal window for effective intervention, which begins sometime (days in animals) after the trauma, perhaps because the stress is very high, and ends when the memory is consolidated and resilient to interference. In fact, changing the timing and number of interventions successfully disrupts the highly traumatic IA memories (Taubenfeld *et al.*, 2009) (Figure 5.6). Furthermore, we found that one or two treatments are sufficient to maximally disrupt the memory, and that the treatment selectively targets the reactivated memory without interfering with the retention of another non-reactivated

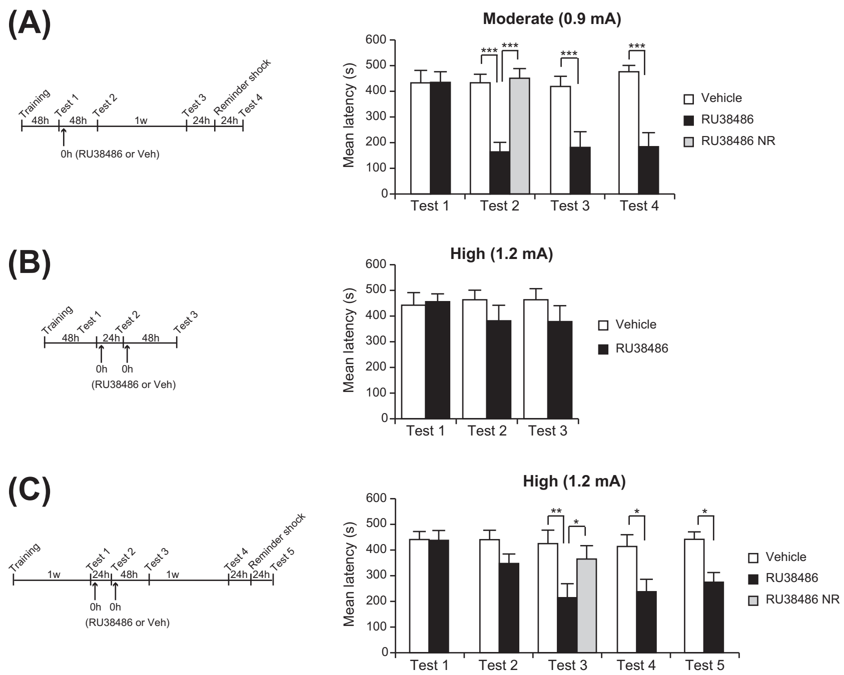


FIGURE 5.6 Preclinical assessment for disrupting IA memory via post-retrieval inhibition of glucocorticoid receptors with systemic injection of RU38486, a glucocorticoid receptor antagonist. Experimental schedules are shown beside each graph. (A) IA memory in rats trained with a moderate (0.9 mA) footshock is disrupted by systemic RU38486 injection given immediately after retrieval (test 1). The retention latencies in the rats that received post-retrieval RU38486 (30 mg/kg body weight) were significantly lower 48 hr later (test 2), 1 week later (test 3), and after a reminder shock (test 4), compared to the vehicle-injected group. NR, rats that received RU38486 in the absence of reactivation. (B) IA memory in rats trained with high-intensity footshock (1.2 mA) is not affected by two treatments consisting of post-retrieval RU38486 injection (30 mg/kg body weight) given 48 hr after training and 1 day later, and tested 48 hr after the last injection. (C) Extending the time between training and memory reactivation allows high-intensity traumatic memory to be disrupted by post-retrieval RU38486 injection. The experiment schedule was similar to that in panel B, except that the first reactivation started at 1 week after IA training. Rats that received two subsequent post-retrieval RU38486 injection had significantly lower retention latencies in subsequent tests compared to the vehicle-injected rats or rats that received RU38486 in the absence of memory retrieval. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. *Source*: Reprinted with permission from Taubenfeld *et al.* (2009).

memory (Taubenfeld *et al.*, 2009). In light of these results, the glucocorticoid pathway appears to be a promising site of pharmacologic intervention for trauma-related pathologies, including PTSD. Another compound that has been employed as a successful pharmacological interference of traumatic memory reconsolidation in both animal fear conditioning (Blundell, Kouser, & Powell, 2008; Helmstetter, Parsons, & Gafford, 2008) and human PTSD is rapamycin (Sirolimus). Interestingly, it appears that more recently developed PTSD may be more amenable to treatment with rapamycin (Suris *et al.*, 2012).

A novel and very interesting alternative to pharmacological disruption, which uses a behavioral design, has recently been provided by animal and human studies based on sequential retrieval (i.e., reconsolidation) and extinction. Extinction training after retrieval of a contextually conditioned fear memory leads to a long-lasting memory reduction if it is presented within the post-retrieval reconsolidation temporal window (Monfils, Cowansage, Klann, & LeDoux, 2009; Schiller *et al.*, 2010) (see Chapters 8 and 9). Further studies should be able to elucidate underlying mechanisms of this interesting approach.

Another pathology for which treatment may take advantage of reconsolidation studies is addiction. Substance abuse generally leads to a chronic condition believed to result from an addict's inability to permanently abstain from drug use. Drug addicts repeatedly relapse to drug seeking even after years of abstinence. This pathologic behavior is frequently induced by the recall of memories and environmental stimuli that are intimately connected to the rewarding effects of the drug (O'Brien, Childress, McLellan, & Ehrman, 1992). Therefore, disruption of memory reconsolidation provides an unprecedented potential strategy to weaken or eliminate memories that facilitate drug addiction. Promising results have been achieved in animals dependent on morphine or cocaine by injecting, after memory reactivation, inhibitors of protein synthesis, extracellular signal-regulated kinase (ERK), β-adrenergic receptors, or matrix metalloproteinases—or by disrupting the expression of the immediate early gene Zif268 either peripherally or within specific brain regions, such as the amygdala, hippocampus, or nucleus accumbens (Brown *et al.*, 2007; Fricks-Gleason, & Marshall, 2008; Lee, Di Ciano, Thomas, & Everitt, 2005; Milekic, Brown, Castellini, & Alberini, 2006; Miller & Marshall, 2005; Robinson & Franklin, 2010; Sorg, 2012; Valjent, Corbillé, Bertran-Gonzalez, Hervé, & Girault, 2006).

In some of these studies, inhibitors were injected in animals that had acquired a place preference in response to the drug of abuse, a learning known as conditioned place preference. Animals learned to associate euphoria of the drug with a specific location, choosing to spend more time in that location. Administration of several of these inhibitors after reactivation of the drug-related memory interfered with its reconsolidation and abolished the place preference. Other studies investigated a different type of task in which animals form a CS−drug association during drug self-administration training, a model of drug seeking (Lee *et al.*, 2005; Lee, Milton, & Everitt, 2006).

These studies particularly showed that infusion of Zif268 antisense oligodeoxynucleotides into the basolateral amygdala, prior to the reactivation of a CS–cocaine association, abolished its impact on the learning of a new cocaine-seeking response or on the maintenance of cocaine seeking, as well as relapse to a previously established drug-seeking behavior. Furthermore, the same group later demonstrated that conditioned withdrawal could be disrupted following reactivation of a CS-withdrawal association (Hellemans, Everitt, & Lee, 2006).

We have also found that disrupting the reconsolidation of a conditioned place preference induced in rats by morphine also leads to a loss of motivational withdrawal evoked in the same place. Interestingly, the hippocampus has a critical role in linking the place preference memory to the context-conditioned withdrawal because interfering with hippocampal molecular mechanisms after the reactivation of morphine conditioned place preference significantly weakens the motivational withdrawal. Thus, targeting the reconsolidation of memories induced by drugs of abuse may prove to be an important strategy for attenuating context-conditioned withdrawal and relapse in opiate addicts (Taubenfeld, Muravieva, Garcia-Osta, & Alberini, 2010). Notably, a protocol of design similar to those employed by Monfils *et al.* (2009) and Schiller *et al.* (2010) has been reported to be successful in disrupting conditioned craving in recovering heroin addicts (Xue *et al.*, 2012).

From all these investigations, it emerges that agents or strategies that disrupt memory reconsolidation represent potentially important approaches for developing novel treatments aimed at weakening pathogenic memories. It is important that future studies determine precisely what response is affected by the post-reactivation treatments, as well as how the age of a memory changes its sensitivity to treatments.

5.7 A MODEL FOR MEMORY RECONSOLIDATION IN HIPPOCAMPAL-DEPENDENT MEMORIES

Memories of a single event that become very long-lasting are evoked by the experience of a salient or emotional event. In contrast, emotionally weaker memories require multiple training trials (i.e., incremental learning) to become equally long-lasting. Let's consider the consolidation and reconsolidation of a long-term memory evoked by a strongly emotional event. The formation of this memory will be processed by the medial temporal lobe through molecular, cellular, and circuitry changes that evolve over an extended time (i.e., weeks in rats and months to years in humans), together with changes occurring in the amygdala. An initial phase of consolidation takes place during the first 1 or 2 days after learning. During this phase, memory is disrupted by treatments that interfere with the synthesis of a number of proteins; however, after this time, it becomes resistant to the same amnesic treatments. This may be interpreted as indicating that consolidation is completed. However, the memory still lies, for some time, in a sensitive, critical phase, during which it can again return to a labile state if reactivated, for example, by retrievals (retraining or reminders), and while in this fragile state, its retention can be bidirectionally modulated. This is the phase of memory reconsolidation. With the passage of time, a gradient of memory stabilization sets in, along with increased resistance to post-retrieval interference. This may be the result of the redistribution of the memory trace and/or the change in the type of trace that underlies the memory.

How does the passage of time contribute to memory strengthening and consolidation? It is possible that the memory trace strengthens because it undergoes implicit reactivations (thus reconsolidations), perhaps as a consequence of constitutive brain activity, such as circadian rhythms or sleep (Diekelmann & Born, 2010; Dudai, 2012; Stickgold *et al.*, 2001). A salient, aversive or traumatic event is also frequently recalled over and over, especially during the first days or weeks (Rubin, Boals, & Berntsen, 2008). Cues often trigger the retrieval of aversive or traumatic experiences. Perhaps these reactivations, both implicit and explicit, serve the biologically important function of consolidating an aversive (important) memory without repeating the aversive experience. Thus, we suggest that through retrieval or reactivation, reconsolidation strengthens or consolidates important memories. Similarly, multiple trials of weaker learning events would produce the same outcome through rounds of consolidations that occur during incremental learning experiences. Hence, for hippocampal-dependent memories, we suggest that reconsolidation is a phase of strengthening that contributes to system memory consolidation.

Reactivation of the memory trace also has another important function, which is to update memories and add new information to the past experience. If the information is a repetition of a previous experience, its reactivations may overlap and perhaps even override the network of the original trace and ultimately lead to strengthening of the same behavior. If the new information encountered is novel and distinct from the recalled experience, the new trace may become linked to the reactivated trace and create a new, parallel memory that coexists with the original one. Importantly, as this new memory is established via a new consolidation process, new associations can be made even when the old reactivated memory is insensitive to post-reactivation interference (Morris *et al.*, 2006; Rodriguez-Ortiz *et al.*, 2005; Winters *et al.*, 2009). Hence, trace activation and reactivation likely create a complex network of stored experiences that is highly dynamic, thus allowing us to adapt to the changing environment.

A better understanding of the dynamics of memory consolidation, reconsolidation, and forgetting should help in the development of novel strategies that can either weaken pathogenic memories of memory components or, when advantageous, enhance memory strength.

ACKNOWLEDGMENTS

This work was supported by grants from the National Institute of Mental Health (R01 MH074736, R01 MH065635), National Institute of Drugs of Abuse (R21 CEBRA DA017672), and the Hirschl, NARSAD, and Philoctetes Foundations to Cristina M. Alberini. We also thank all the members of the Alberini laboratory for their invaluable contributions to the work discussed in this chapter and for their helpful feedback on the manuscript.

REFERENCES

Abrari, K., Rashidy-Pour, A., Semnanian, S., & Fathollahi, Y. (2008). Administration of corticosterone after memory reactivation disrupts subsequent retrieval of a contextual conditioned fear memory: dependence upon training intensity. *Neurobiology of Learning and Memory, 89*(2), 178–184. http://dx.doi.org/10.1016/j.nlm.2007.07.005.

Alberini, C. M. (2005). Mechanisms of memory stabilization: are consolidation and reconsolidation similar or distinct processes? *Trends in Neurosciences, 28*(1), 51–56. http://dx.doi.org/10.1016/j.tins.2004.11.001.

Alberini, C. M. (2008). The role of protein synthesis during the labile phases of memory: revisiting the skepticism. *Neurobiology of Learning and Memory, 89*(3), 234–246. http://dx.doi.org/10.1016/j.nlm.2007.08.007.

Alberini, C. M. (2009). Transcription factors in long-term memory and synaptic plasticity. *Physiological Reviews, 89*(1), 121–145. http://dx.doi.org/10.1152/physrev.00017.2008.

Alberini, C. M. (2011). The role of reconsolidation and the dynamic process of long-term memory formation and storage. *Frontiers in Behavioral Neuroscience, 5*, 12. http://dx.doi.org/10.3389/fnbeh.2011.00012.

Alberini, C. M., Bambah-Mukku, D., & Chen, D. Y. (2012). Memory consolidation and its underlying mechanisms. In K. P. Giese (Ed.), *Memory mechanisms in health and disease*. London: World Scientific Publishing.

Ambrogi Lorenzini, C. G., Baldi, E., Bucherelli, C., Sacchetti, B., & Tassoni, G. (1997). Role of ventral hippocampus in acquisition, consolidation and retrieval of rat's passive avoidance response memory trace. *Brain Research, 768*(1-2), 242–248, doi: S0006-8993(97)00651-3.

Ambrogi Lorenzini, C. G., Baldi, E., Bucherelli, C., Sacchetti, B., & Tassoni, G. (1999). Neural topography and chronology of memory consolidation: a review of functional inactivation findings. *Neurobiology of Learning and Memory, 71*(1), 1–18, S1074-7427(98)93865-5.

Arguello, A. A., Ye, X., Bozdagi, O., Pollonini, G., Tronel, S., Bambah-Mukku, D., Huntley, G. W., Platano, D., & Alberini, C. M. (2013). CCAAT enhancer binding protein delta plays an essential role in memory consolidation and reconsolidation. *Journal of Neuroscience* (in press).

Berman, D. E., & Dudai, Y. (2001). Memory extinction, learning anew, and learning the new: dissociations in the molecular machinery of learning in cortex. *Science, 291*(5512), 2417–2419. http://dx.doi.org/10.1126/science.1058165.

Bernabeu, R., Bevilaqua, L., Ardenghi, P., Bromberg, E., Schmitz, P., Bianchin, M., & Medina, J. H. (1997). Involvement of hippocampal cAMP/cAMP-dependent protein kinase signaling pathways in a late memory consolidation phase of aversively motivated learning in rats. *Proceedings of the National Academy of Sciences of the United States of America, 94*(13), 7041–7046.

Bernabeu, R., Cammarota, M., Izquierdo, I., & Medina, J. H. (1997). Involvement of hippocampal AMPA glutamate receptor changes and the cAMP/protein kinase A/CREB-P signalling pathway in memory consolidation of an avoidance task in rats. *Brazilian Journal of Medical and Biological Research, 30*(8), 961–965.

Bindra, D., & Cameron, L. (1953). Changes in experimentally produced anxiety with the passage of time: incubation effect. *Journal of Experimental Psychology, 45*(3), 197–203.

Bizon, J. L., Lasarge, C. L., Montgomery, K. S., McDermott, A. N., Setlow, B., & Griffith, W. H. (2009). Spatial reference and working memory across the lifespan of male Fischer 344 rats. *Neurobiology of Aging, 30*(4), 646–655. http://dx.doi.org/10.1016/j.neurobiolaging.2007.08.004.

Blundell, J., Kouser, M., & Powell, C. M. (2008). Systemic inhibition of mammalian target of rapamycin inhibits fear memory reconsolidation. *Neurobiology of Learning and Memory, 90*(1), 28–35. http://dx.doi.org/10.1016/j.nlm.2007.12.004.

Boccia, M. M., Blake, M. G., Acosta, G. B., & Baratti, C. M. (2006). Post-retrieval effects of icv infusions of hemicholinium in mice are dependent on the age of the original memory. *Learning & Memory, 13*(3), 376–381, 13/3/376 10.1101/lm.150306.

Bonini, J. S., Cammarota, M., Kerr, D. S., Bevilaqua, L. R., & Izquierdo, I. (2005). Inhibition of PKC in basolateral amygdala and posterior parietal cortex impairs consolidation of inhibitory avoidance memory. *Pharmacology, Biochemistry, & Behavior, 80*(1), 63−67, S0091-3057(04)00332-6.

Bonini, J. S., Rodrigues, L., Kerr, D. S., Bevilaqua, L. R., Cammarota, M., & Izquierdo, I. (2003). AMPA/kainate and group-I metabotropic receptor antagonists infused into different brain areas impair memory formation of inhibitory avoidance in rats. *Behavioural Pharmacology, 14*(2), 161−166. http://dx.doi.org/10.1097/01.fbp.0000063621.43827.8c.

Born, J., & Wilhelm, I. (2012). System consolidation of memory during sleep. *Psychological Research, 76*(2), 192−203. http://dx.doi.org/10.1007/s00426-011-0335-6.

Brightwell, J. J., Gallagher, M., & Colombo, P. J. (2004). Hippocampal CREB1 but not CREB2 is decreased in aged rats with spatial memory impairments. *Neurobiology of Learning and Memory, 81*(1), 19−26.

Brown, T. E., Forquer, M. R., Cocking, D. L., Jansen, H. T., Harding, J. W., & Sorg, B. A. (2007). Role of matrix metalloproteinases in the acquisition and reconsolidation of cocaine-induced conditioned place preference. *Learning & Memory, 14*(3), 214−223. http://dx.doi.org/10.1101/lm.476207.

Burke, S. N., & Barnes, C. A. (2006). Neural plasticity in the ageing brain. *Nature Reviews Neuroscience, 7*(1), 30−40. http://dx.doi.org/10.1038/nrn1809.

Bustos, S. G., Maldonado, H., & Molina, V. A. (2009). Disruptive effect of midazolam on fear memory reconsolidation: decisive influence of reactivation time span and memory age. *Neuropsychopharmacology, 34*(2), 446−457, npp200875.

Cai, W.-H., Blundell, J., Han, J., Greene, R. W., & Powell, C. M. (2006). Postreactivation glucocorticoids impair recall of established fear memory. *Journal of Neuroscience, 26*(37), 9560−9566. http://dx.doi.org/10.1523/JNEUROSCI.2397-06.2006.

Canal, C. E., Chang, Q., & Gold, P. E. (2008). Intra-amygdala injections of CREB antisense impair inhibitory avoidance memory: role of norepinephrine and acetylcholine. *Learning & Memory, 15*(9), 677−686. http://dx.doi.org/10.1101/15/9/677, [pii] 10.1101/lm.904308.

Cansino, S. (2009). Episodic memory decay along the adult lifespan: a review of behavioral and neurophysiological evidence. *International Journal of Psychophysiology, 71*(1), 64−69. http://dx.doi.org/10.1016/j.ijpsycho.2008.07.005.

Carbo Tano, M., Molina, V. A., Maldonado, H., & Pedreira, M. E. (2009). Memory consolidation and reconsolidation in an invertebrate model: the role of the GABAergic system. *Neuroscience, 158*(2), 387−401. http://dx.doi.org/10.1016/S0306-4522(08)01591-1, [pii] 10.1016/j.neuroscience.2008.10.039.

Chen, D. Y., Bambah-Mukku, D., Pollonini, G., & Alberini, C. M. (2012). Glucocorticoid receptors recruit the CaMKIIα, BDNF-CREB pathways to mediate memory consolidation. *Nature Neuroscience, 15*(12), 1707−1714. http://dx.doi.org/10.1038/nn.3266.

Chen, D. Y., Stern, S. A., Garcia-Osta, A., Saunier-Rebori, B., Pollonini, G., Bambah-Mukku, D., & Alberini, C. M. (2011). A critical role for IGF-II in memory consolidation and enhancement. *Nature, 469*(7331), 491−497. http://dx.doi.org/10.1038/nature09667, [pii] 10.1038/nature09667.

Countryman, R. A., & Gold, P. E. (2007). Rapid forgetting of social transmission of food preferences in aged rats: relationship to hippocampal CREB activation. *Learning & Memory, 14*(5), 350−358. http://dx.doi.org/10.1101/lm.524907.

Davis, H. P., & Squire, L. R. (1984). Protein synthesis and memory: a review. *Psychological Bulletin, 96*(3), 518−559.

de Souza, F. A., Sanchis-Segura, C., Fukada, S. Y., de Bortoli, V. C., Zangrossi, H., Jr., & de Oliveira, A. M. (2004). Intracerebroventricular effects of angiotensin II on a step-through passive avoidance task in rats. *Neurobiology of Learning and Memory, 81*(1), 100−103, S1074742703000959.

Dębiec, J., Doyère, V., Nader, K., & LeDoux, J. E. (2006). Directly reactivated, but not indirectly reactivated, memories undergo reconsolidation in the amygdala. *Proceedings of the National Academy of Sciences of the United States of America, 103*(9), 3428−3433. http://dx.doi.org/10.1073/pnas.0507168103.

Dębiec, J., & LeDoux, J. E. (2004). Disruption of reconsolidation but not consolidation of auditory fear conditioning by noradrenergic blockade in the amygdala. *Neuroscience, 129*(2), 267–272. http://dx.doi.org/10.1016/j.neuroscience.2004.08.018.

Dębiec, J., LeDoux, J. E., & Nader, K. (2002). Cellular and systems reconsolidation in the hippocampus. *Neuron, 36*(3), 527–538.

Devauges, V., & Sara, S. J. (1991). Memory retrieval enhancement by locus coeruleus stimulation: evidence for mediation by beta-receptors. *Behavioural Brain Research, 43*(1), 93–97.

DeVietti, T. L., Conger, G. L., & Kirkpatrick, B. R. (1977). Comparison of the enhancement gradients of retention obtained with stimulation of the mesencephalic reticular formation after training or memory reactivation. *Physiology & Behavior, 19*(4), 549–554.

Diekelmann, S., & Born, J. (2010). The memory function of sleep. *Nature Reviews Neuroscience, 11*(2), 114–126. http://dx.doi.org/10.1038/nrn2762.

Diekelmann, S., Büchel, C., Born, J., & Rasch, B. (2011). Labile or stable: opposing consequences for memory when reactivated during waking and sleep. *Nature Neuroscience, 14*(3), 381–386. http://dx.doi.org/10.1038/nn.2744.

Diergaarde, L., Schoffelmeer, A. N., & De Vries, T. J. (2006). Beta-adrenoceptor mediated inhibition of long-term reward-related memory reconsolidation. *Behavioural Brain Research, 170*(2), 333–336. http://dx.doi.org/10.1016/j.bbr.2006.02.014.

Dudai, Y. (1996). Consolidation: fragility on the road to the engram. *Neuron, 17*(3), 367–370.

Dudai, Y. (2004). The neurobiology of consolidations, or, how stable is the engram? *Annual Review of Psychology, 55*, 51–86. http://dx.doi.org/10.1146/annurev.psych.55.090902.142050.

Dudai, Y. (2012). The restless engram: consolidations never end. *Annual Review of Neuroscience, 35*, 227–247. http://dx.doi.org/10.1146/annurev-neuro-062111-150500.

Dudai, Y., & Eisenberg, M. (2004). Rites of passage of the engram: reconsolidation and the lingering consolidation hypothesis. *Neuron, 44*(1), 93–100. http://dx.doi.org/10.1016/j.neuron.2004.09.003.

Eisenberg, M., & Dudai, Y. (2004). Reconsolidation of fresh, remote, and extinguished fear memory in Medaka: old fears don't die. *European Journal of Neuroscience, 20*(12), 3397–3403. http://dx.doi.org/10.1111/j.1460-9568.2004.03818.x.

Eisenberg, M., Kobilo, T., Berman, D. E., & Dudai, Y. (2003). Stability of retrieved memory: inverse correlation with trace dominance. *Science, 301*(5636), 1102–1104. http://dx.doi.org/10.1126/science.1086881 301/5636/1102.

Fischer, A., Sananbenesi, F., Schrick, C., Spiess, J., & Radulovic, J. (2004). Distinct roles of hippocampal *de novo* protein synthesis and actin rearrangement in extinction of contextual fear. *Journal of Neuroscience, 24*(8), 1962–1966. http://dx.doi.org/10.1523/JNEUROSCI.5112-03.2004.

Forcato, C., Rodríguez, M. L. C., Pedreira, M. E., & Maldonado, H. (2010). Reconsolidation in humans opens up declarative memory to the entrance of new information. *Neurobiology of Learning and Memory, 93*(1), 77–84. http://dx.doi.org/10.1016/j.nlm.2009.08.006.

Frankland, P. W., & Bontempi, B. (2005). The organization of recent and remote memories. *Nature Reviews Neuroscience, 6*(2), 119–130. http://dx.doi.org/10.1038/nrn1607.

Frankland, P. W., Ding, H. K., Takahashi, E., Suzuki, A., Kida, S., & Silva, A. J. (2006). Stability of recent and remote contextual fear memory. *Learning & Memory, 13*(4), 451–457. http://dx.doi.org/10.1101/lm.183406.

Frenkel, L., Maldonado, H., & Delorenzi, A. (2005). Memory strengthening by a real-life episode during reconsolidation: an outcome of water deprivation via brain angiotensin II. *European Journal of Neuroscience, 22*(7), 1757–1766. http://dx.doi.org/10.1111/EJN4373, [pii] 10.1111/j.1460-9568.2005.04373.x.

Fricks-Gleason, A. N., & Marshall, J. F. (2008). Post-retrieval beta-adrenergic receptor blockade: effects on extinction and reconsolidation of cocaine-cue memories. *Learning & Memory, 15*(9), 643–648. http://dx.doi.org/10.1101/lm.1054608.

Gallagher, M., & Rapp, P. R. (1997). The use of animal models to study the effects of aging on cognition. *Annual Review of Psychology, 48*, 339–370. http://dx.doi.org/10.1146/annurev.psych.48.1.339.

Garcia-Osta, A., Tsokas, P., Pollonini, G., Landau, E. M., Blitzer, R., & Alberini, C. M. (2006). MuSK expressed in the brain mediates cholinergic responses, synaptic plasticity, and memory formation. *Journal of Neuroscience, 26*(30), 7919–7932, 26/30/7919 10.1523/JNEUROSCI.1674-06.2006.

Graves, L., Pack, A., & Abel, T. (2001). Sleep and memory: a molecular perspective. *Trends in Neurosciences, 24*(4), 237–243.

Grimm, J. W., Fyall, A. M., & Osincup, D. P. (2005). Incubation of sucrose craving: effects of reduced training and sucrose pre-loading. *Physiology & Behavior, 84*(1), 73–79. http://dx.doi.org/10.1016/j.physbeh.2004.10.011.

Gusev, P. A., & Gubin, A. N. (2010a). Arc/Arg3.1 mRNA global expression patterns elicited by memory recall in cerebral cortex differ for remote versus recent spatial memories. *Frontiers in Integrative Neuroscience, 4*, 15. http://dx.doi.org/10.3389/fnint.2010.00015.

Gusev, P. A., & Gubin, A. N. (2010b). Recent and remote memory recalls modulate different sets of stereotypical interlaminar correlations in Arc/Arg3.1 mRNA expression in cortical areas. *Brain Research, 1352*, 118–139. http://dx.doi.org/10.1016/S0006-8993(10)01490-3, [pii] 10.1016/j.brainres.2010.06.064.

Hellemans, K. G. C., Everitt, B. J., & Lee, J. L. C. (2006). Disrupting reconsolidation of conditioned withdrawal memories in the basolateral amygdala reduces suppression of heroin seeking in rats. *Journal of Neuroscience, 26*(49), 12694–12699. http://dx.doi.org/10.1523/JNEUROSCI.3101-06.2006.

Helmstetter, F. J., Parsons, R. G., & Gafford, G. M. (2008). Macromolecular synthesis, distributed synaptic plasticity, and fear conditioning. *Neurobiology of Learning and Memory, 89*(3), 324–337. http://dx.doi.org/10.1016/j.nlm.2007.09.002.

Hoeffer, C. A., Cowansage, K. K., Arnold, E. C., Banko, J. L., Moerke, N. J., Rodriguez, R., & Klann, E. (2011). Inhibition of the interactions between eukaryotic initiation factors 4E and 4G impairs long-term associative memory consolidation but not reconsolidation. *Proceedings of the National Academy of Sciences of the United States of America, 108*(8), 3383–3388, 1013063108 10.1073/pnas.1013063108.

Impey, S., Smith, D. M., Obrietan, K., Donahue, R., Wade, C., & Storm, D. R. (1998). Stimulation of cAMP response element (CRE)-mediated transcription during contextual learning. *Nature Neuroscience, 1*(7), 595–601. http://dx.doi.org/10.1038/2830.

Inda, M. C., Muravieva, E. V., & Alberini, C. M. (2011). Memory retrieval and the passage of time: from reconsolidation and strengthening to extinction. *Journal of Neuroscience, 31*(5), 1635–1643. http://dx.doi.org/10.1523/JNEUROSCI.4736-10.2011.

Izquierdo, I., Quillfeldt, J. A., Zanatta, M. S., Quevedo, J., Schaeffer, E., Schmitz, P. K., & Medina, J. H. (1997). Sequential role of hippocampus and amygdala, entorhinal cortex and parietal cortex in formation and retrieval of memory for inhibitory avoidance in rats. *European Journal of Neuroscience, 9*(4), 786–793.

Jerusalinsky, D., Ferreira, M. B., Walz, R., Da Silva, R. C., Bianchin, M., Ruschel, A. C., & Izquierdo, I. (1992). Amnesia by post-training infusion of glutamate receptor antagonists into the amygdala, hippocampus, and entorhinal cortex. *Behavioral and Neural Biology, 58*(1), 76–80.

Jobim, P. F., Pedroso, T. R., Christoff, R. R., Werenicz, A., Maurmann, N., Reolon, G. K., & Roesler, R. (2012). Inhibition of mTOR by rapamycin in the amygdala or hippocampus impairs formation and reconsolidation of inhibitory avoidance memory. *Neurobiology of Learning and Memory, 97*(1), 105–112. http://dx.doi.org/10.1016/S1074-7427(11)00176-6, [pii] 10.1016/j.nlm.2011.10.002.

Kim, J. J., & Fanselow, M. S. (1992). Modality-specific retrograde amnesia of fear. *Science, 256*(5057), 675–677.

Kindt, M., Soeter, M., & Vervliet, B. (2009). Beyond extinction: erasing human fear responses and preventing the return of fear. *Nature Neuroscience, 12*(3), 256–258. http://dx.doi.org/10.1038/nn.2271.

Kitamura, T., Saitoh, Y., Takashima, N., Murayama, A., Niibori, Y., Ageta, H., & Inokuchi, K. (2009). Adult neurogenesis modulates the hippocampus-dependent period of associative fear memory. *Cell, 139*(4), 814–827. http://dx.doi.org/10.1016/S0092-8674(09)01309-9, [pii] 10.1016/j.cell.2009.10.020.

LaSarge, C. L., Montgomery, K. S., Tucker, C., Slaton, G. S., Griffith, W. H., Setlow, B., & Bizon, J. L. (2007). Deficits across multiple cognitive domains in a subset of aged Fischer 344 rats. *Neurobiology of Aging, 28*(6), 928–936. http://dx.doi.org/10.1016/j.neurobiolaging.2006.04.010.

Lee, J. L. C., Everitt, B. J., & Thomas, K. L. (2004). Independent cellular processes for hippocampal memory consolidation and reconsolidation. *Science, 304*(5672), 839–843. http://dx.doi.org/10.1126/science.1095760 1095760.

Lee, J. L. C., Di Ciano, P., Thomas, K. L., & Everitt, B. J. (2005). Disrupting reconsolidation of drug memories reduces cocaine-seeking behavior. *Neuron, 47*(6), 795–801. http://dx.doi.org/10.1016/j.neuron.2005.08.007.

Lee, J. L. C., Milton, A. L., & Everitt, B. J. (2006). Cue-induced cocaine seeking and relapse are reduced by disruption of drug memory reconsolidation. *Journal of Neuroscience, 26*(22), 5881–5887. http://dx.doi.org/10.1523/JNEUROSCI.0323-06.2006.

Lewis, D. J. (1979). Psychobiology of active and inactive memory. *Psychological Bulletin, 86*(5), 1054–1083.

Litvin, O. O., & Anokhin, K. V. (2000). Mechanisms of memory reorganization during retrieval of acquired behavioral experience in chicks: the effects of protein synthesis inhibition in the brain. *Neuroscience and Behavioral Physiology, 30*(6), 671–678.

Lorenzini, C. A., Baldi, E., Bucherelli, C., Sacchetti, B., & Tassoni, G. (1996). Role of dorsal hippocampus in acquisition, consolidation and retrieval of rat's passive avoidance response: a tetrodotoxin functional inactivation study. *Brain Research, 730*(1-2), 32–39.

Lu, L., Grimm, J. W., Hope, B. T., & Shaham, Y. (2004). Incubation of cocaine craving after withdrawal: a review of preclinical data. *Neuropharmacology, 47*(Suppl. 1), 214–226. http://dx.doi.org/10.1016/j.neuropharm.2004.06.027.

Luft, A. R., Buitrago, M. M., Kaelin-Lang, A., Dichgans, J., & Schulz, J. B. (2004). Protein synthesis inhibition blocks consolidation of an acrobatic motor skill. *Learning & Memory, 11*(4), 379–382. http://dx.doi.org/10.1101/lm.72604.

Luft, A. R., Buitrago, M. M., Ringer, T., Dichgans, J., & Schulz, J. B. (2004). Motor skill learning depends on protein synthesis in motor cortex after training. *Journal of Neuroscience, 24*(29), 6515–6520. http://dx.doi.org/10.1523/JNEUROSCI.1034-04.2004.

Malin, E. L., Ibrahim, D. Y., Tu, J. W., & McGaugh, J. L. (2007). Involvement of the rostral anterior cingulate cortex in consolidation of inhibitory avoidance memory: interaction with the basolateral amygdala. *Neurobiology of Learning and Memory, 87*(2), 295–302. http://dx.doi.org/10.1016/S1074-7427(06)00134-1, [pii] 10.1016/j.nlm.2006.09.004.

Marshall, L., Helgadottir, H., Molle, M., & Born, J. (2006). Boosting slow oscillations during sleep potentiates memory. *Nature, 444*(7119), 610–613. http://dx.doi.org/10.1038/nature05278.

McGaugh, J. L. (2000). Memory—A century of consolidation. *Science, 287*(5451), 248–251.

McGaugh, J. L. (2004). The amygdala modulates the consolidation of memories of emotionally arousing experiences. *Annual Review of Neuroscience, 27*, 1–28. http://dx.doi.org/10.1146/annurev.neuro.27.070203.144157.

McGaugh, J. L., & Roozendaal, B. (2002). Role of adrenal stress hormones in forming lasting memories in the brain. *Current Opinion in Neurobiology, 12*(2), 205–210.

McIntyre, C. K., Power, A. E., Roozendaal, B., & McGaugh, J. L. (2003). Role of the basolateral amygdala in memory consolidation. *Annals of the New York Academy of Sciences, 985*, 273–293.

Meiri, N., & Rosenblum, K. (1998). Lateral ventricle injection of the protein synthesis inhibitor anisomycin impairs long-term memory in a spatial memory task. *Brain Research, 789*(1), 48–55.

Milekic, M. H., & Alberini, C. M. (2002). Temporally graded requirement for protein synthesis following memory reactivation. *Neuron, 36*(3), 521–525, S0896627302009765.

Milekic, M. H., Brown, S. D., Castellini, C., & Alberini, C. M. (2006). Persistent disruption of an established morphine conditioned place preference. *Journal of Neuroscience, 26*(11), 3010–3020. http://dx.doi.org/10.1523/JNEUROSCI.4818-05.2006.

Milekic, M. H., Pollonini, G., & Alberini, C. M. (2007). Temporal requirement of C/EBPbeta in the amygdala following reactivation but not acquisition of inhibitory avoidance. *Learning & Memory, 14*(7), 504–511. http://dx.doi.org/10.1101/14/7/504, [pii] 10.1101/lm.598307.

Miller, C. A., & Marshall, J. F. (2005). Molecular substrates for retrieval and reconsolidation of cocaine-associated contextual memory. *Neuron, 47*(6), 873–884. http://dx.doi.org/10.1016/j.neuron.2005.08.006.

Milton, A. L., Lee, J. L., & Everitt, B. J. (2008). Reconsolidation of appetitive memories for both natural and drug reinforcement is dependent on β-adrenergic receptors. *Learning & Memory, 15*(2), 88–92. http://dx.doi.org/10.1101/lm.825008.

Molle, M., Yeshenko, O., Marshall, L., Sara, S. J., & Born, J. (2006). Hippocampal sharp wave-ripples linked to slow oscillations in rat slow-wave sleep. *Journal of Neurophysiology, 96*(1), 62–70. http://dx.doi.org/10.1152/jn.00014.2006.

Monfils, M.-H., Cowansage, K. K., Klann, E., & LeDoux, J. E. (2009). Extinction-reconsolidation boundaries: key to persistent attenuation of fear memories. *Science, 324*(5929), 951–955. http://dx.doi.org/10.1126/science.1167975.

Monti, B., Berteotti, C., & Contestabile, A. (2005). Dysregulation of memory-related proteins in the hippocampus of aged rats and their relation with cognitive impairment. *Hippocampus, 15*(8), 1041–1049. http://dx.doi.org/10.1002/hipo.20099.

Morris, R. G. M., Inglis, J., Ainge, J. A., Olverman, H. J., Tulloch, J., Dudai, Y., & Kelly, P. A. T. (2006). Memory reconsolidation: sensitivity of spatial memory to inhibition of protein synthesis in dorsal hippocampus during encoding and retrieval. *Neuron, 50*(3), 479–489. http://dx.doi.org/10.1016/j.neuron.2006.04.012.

Moscovitch, M., Rosenbaum, R. S., Gilboa, A., Addis, D. R., Westmacott, R., Grady, C., & Nadel, L. (2005). Functional neuroanatomy of remote episodic, semantic and spatial memory: a unified account based on multiple trace theory. *Journal of Anatomy, 207*(1), 35–66. http://dx.doi.org/10.1111/JOA421, [pii] 10.1111/j.1469-7580.2005.00421.x.

Muravieva, E. V., & Alberini, C. M. (2010). Limited efficacy of propranolol on the reconsolidation of fear memories. *Learning & Memory, 17*(6), 306–313. http://dx.doi.org/10.1101/lm.1794710.

Nader, K., & Einarsson, E. O. (2010). Memory reconsolidation: an update. *Annals of the New York Academy of Sciences, 1191*, 27–41. http://dx.doi.org/10.1111/NYAS5443, [pii] 10.1111/j.1749-6632.2010.05443.x.

Nader, K., & Hardt, O. (2009). A single standard for memory: the case for reconsolidation. *Nature Reviews Neuroscience, 10*(3), 224–234. http://dx.doi.org/10.1038/nrn2590.

Nader, K., Schafe, G. E., & Le Doux, J. E. (2000). Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval. *Nature, 406*(6797), 722–726. http://dx.doi.org/10.1038/35021052.

Nader, K., Schafe, G. E., & LeDoux, J. E. (2000). The labile nature of consolidation theory. *Nature Reviews Neuroscience, 1*(3), 216–219. http://dx.doi.org/10.1038/35044580.

Nikzad, S., Vafaei, A. A., Rashidy-Pour, A., & Haghighi, S. (2011). Systemic and intrahippocampal administrations of the glucocorticoid receptor antagonist RU38486 impairs fear memory reconsolidation in rats. *Stress, 14*(4), 459–464. http://dx.doi.org/10.3109/10253890.2010.548171.

O'Brien, C. P., Childress, A. R., McLellan, A. T., & Ehrman, R. (1992). Classical conditioning in drug-dependent humans. *Annals of the New York Academy of Sciences, 654*, 400–415.

O'Donnell, T., Hegadoren, K. M., & Coupland, N. C. (2004). Noradrenergic mechanisms in the pathophysiology of post-traumatic stress disorder. *Neuropsychobiology, 50*(4), 273–283. http://dx.doi.org/10.1159/000080952.

Pedreira, M. E., & Maldonado, H. (2003). Protein synthesis subserves reconsolidation or extinction depending on reminder duration. *Neuron, 38*(6), 863–869, S0896627303003520.

Pickens, C. L., Airavaara, M., Theberge, F., Fanous, S., Hope, B. T., & Shaham, Y. (2011). Neurobiology of the incubation of drug craving. *Trends in Neurosciences, 34*(8), 411–420. http://dx.doi.org/10.1016/j.tins.2011.06.001.

Pickens, C. L., Golden, S. A., Adams-Deutsch, T., Nair, S. G., & Shaham, Y. (2009). Long-lasting incubation of conditioned fear in rats. *Biological Psychiatry, 65*(10), 881−886. http://dx.doi.org/10.1016/j.biopsych.2008.12.010.

Pitman, R. K. (1989). Post-traumatic stress disorder, hormones, and memory. *Biological Psychiatry, 26*(3), 221−223.

Pitman, R. K., & Delahanty, D. L. (2005). Conceptually driven pharmacologic approaches to acute trauma. *CNS Spectrums, 10*(2), 99−106.

Power, A. E., Berlau, D. J., McGaugh, J. L., & Steward, O. (2006). Anisomycin infused into the hippocampus fails to block "reconsolidation" but impairs extinction: The role of re-exposure duration. *Learning & Memory, 13*(1), 27−34, 13/1/27 10.1101/lm.91206.

Przybyslawski, J., Roullet, P., & Sara, S. J. (1999). Attenuation of emotional and nonemotional memories after their reactivation: role of beta adrenergic receptors. *Journal of Neuroscience, 19*(15), 6623−6628.

Quillfeldt, J. A., Zanatta, M. S., Schmitz, P. K., Quevedo, J., Schaeffer, E., Lima, J. B., & Izquierdo, I. (1996). Different brain areas are involved in memory expression at different times from training. *Neurobiology of Learning and Memory, 66*(2), 97−101. http://dx.doi.org/10.1006/S1074-7427(96)90050-7, [pii] 10.1006/nlme.1996.0050.

Restivo, L., Vetere, G., Bontempi, B., & Ammassari-Teule, M. (2009). The formation of recent and remote memory is associated with time-dependent formation of dendritic spines in the hippocampus and anterior cingulate cortex. *Journal of Neuroscience, 29*(25), 8206−8214. http://dx.doi.org/10.1523/29/25/8206, [pii] 10.1523/JNEUROSCI.0966-09.2009.

Robinson, M. J. F., & Franklin, K. B. J. (2010). Reconsolidation of a morphine place preference: impact of the strength and age of memory on disruption by propranolol and midazolam. *Behavioural Brain Research, 213*(2), 201−207. http://dx.doi.org/10.1016/j.bbr.2010.04.056.

Robitsek, R. J., Fortin, N. J., Koh, M. T., Gallagher, M., & Eichenbaum, H. (2008). Cognitive aging: a common decline of episodic recollection and spatial memory in rats. *Journal of Neuroscience, 28*(36), 8945−8954. http://dx.doi.org/10.1523/JNEUROSCI.1893-08.2008.

Rodriguez-Ortiz, C. J., De la Cruz, V., Gutiérrez, R., & Bermudez-Rattoni, F. (2005). Protein synthesis underlies post-retrieval memory consolidation to a restricted degree only when updated information is obtained. *Learning & Memory, 12*(5), 533−537. http://dx.doi.org/10.1101/lm.94505.

Roozendaal, B., Brunson, K. L., Holloway, B. L., McGaugh, J. L., & Baram, T. Z. (2002). Involvement of stress-released corticotropin-releasing hormone in the basolateral amygdala in regulating memory consolidation. *Proceedings of the National Academy of Sciences of the United States of America, 99*(21), 13908−13913. http://dx.doi.org/10.1073/pnas.212504599 212504599.

Ross, R. S., & Eichenbaum, H. (2006). Dynamics of hippocampal and cortical activation during consolidation of a nonspatial memory. *Journal of Neuroscience, 26*(18), 4852−4859, 26/18/4852 10.1523/JNEUROSCI.0659-06.2006.

Rossato, J. I., Bonini, J. S., Coitinho, A. S., Vianna, M. R., Medina, J. H., Cammarota, M., & Izquierdo, I. (2004). Retrograde amnesia induced by drugs acting on different molecular systems. *Behavioural Neuroscience, 118*(3), 563−568. http://dx.doi.org/10.1037/0735-7044.118.3.563.

Rubin, D. C., Boals, A., & Berntsen, D. (2008). Memory in posttraumatic stress disorder: properties of voluntary and involuntary, traumatic and nontraumatic autobiographical memories in people with and without posttraumatic stress disorder symptoms. *Journal of Experimental Psychology. General, 137*(4), 591−614. http://dx.doi.org/10.1037/a0013165.

Sacco, T., & Sacchetti, B. (2010). Role of secondary sensory cortices in emotional memory storage and retrieval in rats. *Science, 329*(5992), 649−656, 329/5992/649 10.1126/science.1183165.

Santini, E., Ge, H., Ren, K., Peña de Ortiz, S., & Quirk, G. J. (2004). Consolidation of fear extinction requires protein synthesis in the medial prefrontal cortex. *Journal of Neuroscience, 24*(25), 5704−5710. http://dx.doi.org/10.1523/JNEUROSCI.0786-04.2004.

Sara, S. J. (2000). Strengthening the shaky trace through retrieval. *Nature Reviews Neuroscience, 1*(3), 212−213. http://dx.doi.org/10.1038/35044575.

Schiller, D., Monfils, M.-H., Raio, C. M., Johnson, D. C., LeDoux, J. E., & Phelps, E. A. (2010). Preventing the return of fear in humans using reconsolidation update mechanisms. *Nature, 463*(7277), 49−53. http://dx.doi.org/10.1038/nature08637.

Sirota, A., Csicsvari, J., Buhl, D., & Buzsaki, G. (2003). Communication between neocortex and hippocampus during sleep in rodents. *Proceedings of the National Academy of Sciences of the United States of America, 100*(4), 2065−2069. http://dx.doi.org/10.1073/pnas.0437938100.

Slipczuk, L., Bekinschtein, P., Katche, C., Cammarota, M., Izquierdo, I., & Medina, J. H. (2009). BDNF activates mTOR to regulate GluR1 expression required for memory formation. *PLoS ONE, 4*(6). http://dx.doi.org/10.1371/journal.pone.0006007. e6007.

Smith, C. N., & Squire, L. R. (2009). Medial temporal lobe activity during retrieval of semantic memory is related to the age of the memory. *Journal of Neuroscience, 29*(4), 930−938, 29/4/930 10.1523/JNEUROSCI.4545-08.2009.

Sorg, B. A. (2012). Reconsolidation of drug memories. *Neuroscience and Biobehavioral Reviews, 36*(5), 1400−1417. http://dx.doi.org/10.1016/j.neubiorev.2012.02.004.

Squire, L. R., & Alvarez, P. (1995). Retrograde amnesia and memory consolidation: a neurobiological perspective. *Current Opinion in Neurobiology, 5*(2), 169−177.

Squire, L. R., Clark, R. E., & Knowlton, B. J. (2001). Retrograde amnesia. *Hippocampus, 11*(1), 50−55. http://dx.doi.org/10.1002/1098-1063(2001)11, 1<50::AID-HIPO1019>3.0.CO;2-G.

Squire, L. R., Stark, C. E., & Clark, R. E. (2004). The medial temporal lobe. *Annual Review of Neuroscience, 27*, 279−306. http://dx.doi.org/10.1146/annurev.neuro.27.070203.144130.

Staddon, J. E., & Cerutti, D. T. (2003). Operant conditioning. *Annual Review of Psychology, 54*, 115−144. http://dx.doi.org/10.1146/annurev.psych.54.101601.145124 101601.145124.

Stickgold, R., Hobson, J. A., Fosse, R., & Fosse, M. (2001). Sleep, learning, and dreams: offline memory reprocessing. *Science, 294*(5544), 1052−1057. http://dx.doi.org/10.1126/science.1063530.

Suárez, L. D., Smal, L., & Delorenzi, A. (2010). Updating contextual information during consolidation as result of a new memory trace. *Neurobiology of Learning and Memory, 93*(4), 561−571. http://dx.doi.org/10.1016/j.nlm.2010.02.004.

Suris, A., Smith, J., Powell, C. M., & North, C. S. (2012). Interfering with the reconsolidation of traumatic memory: sirolimus as a novel agent for treating veterans with posttraumatic stress disorder. *Annals of Clinical Psychiatry.* in press.

Suzuki, A., Josselyn, S. A., Frankland, P. W., Masushige, S., Silva, A. J., & Kida, S. (2004). Memory reconsolidation and extinction have distinct temporal and biochemical signatures. *Journal of Neuroscience, 24*(20), 4787−4795. http://dx.doi.org/10.1523/JNEUROSCI.5491-03.2004.

Takehara-Nishiuchi, K., & McNaughton, B. L. (2008). Spontaneous changes of neocortical code for associative memory during consolidation. *Science, 322*(5903), 960−963. http://dx.doi.org/10.1126/322/5903/960, [pii] 10.1126/science.1161299.

Taubenfeld, S. M., Milekic, M. H., Monti, B., & Alberini, C. M. (2001a). The consolidation of new but not reactivated memory requires hippocampal C/EBPbeta. *Nature Neuroscience, 4*(8), 813−818. http://dx.doi.org/10.1038/90520.

Taubenfeld, S. M., Muravieva, E. V., Garcia-Osta, A., & Alberini, C. M. (2010). Disrupting the memory of places induced by drugs of abuse weakens motivational withdrawal in a context-dependent manner. *Proceedings of the National Academy of Sciences of the United States of America, 107*(27), 12345−12350. http://dx.doi.org/10.1073/pnas.1003152107.

Taubenfeld, S. M., Riceberg, J. S., New, A. S., & Alberini, C. M. (2009). Preclinical assessment for selectively disrupting a traumatic memory via postretrieval inhibition of glucocorticoid receptors. *Biological Psychiatry, 65*(3), 249−257. http://dx.doi.org/10.1016/j.biopsych.2008.07.005.

Taubenfeld, S. M., Wiig, K. A., Monti, B., Dolan, B., Pollonini, G., & Alberini, C. M. (2001b). Fornix-dependent induction of hippocampal CCAAT enhancer-binding protein β and δ co-localizes with phosphorylated cAMP response element-binding protein and accompanies long-term memory consolidation. *Journal of Neuroscience, 21*(1), 84–91, 21/1/84.

Taubenfeld, S. M., Wiig, K. A., Bear, M. F., & Alberini, C. M. (1999). A molecular correlate of memory and amnesia in the hippocampus. *Nature Neuroscience, 2*(4), 309–310. http://dx.doi.org/10.1038/7217.

Teixeira, C. M., Pomedli, S. R., Maei, H. R., Kee, N., & Frankland, P. W. (2006). Involvement of the anterior cingulate cortex in the expression of remote spatial memory. *Journal of Neuroscience, 26*(29), 7555–7564. http://dx.doi.org/10.1523/26/29/7555, [pii] 10.1523/JNEUROSCI.1068-06.2006.

Tollenaar, M. S., Elzinga, B. M., Spinhoven, P., & Everaerd, W. (2009). Immediate and prolonged effects of cortisol, but not propranolol, on memory retrieval in healthy young men. *Neurobiology of Learning and Memory, 91*(1), 23–31. http://dx.doi.org/10.1016/j.nlm.2008.08.002.

Touzani, K., Puthanveettil, S. V., & Kandel, E. R. (2007). Consolidation of learning strategies during spatial working memory task requires protein synthesis in the prefrontal cortex. *Proceedings of the National Academy of Sciences of the United States of America, 104*(13), 5632–5637. http://dx.doi.org/10.1073/pnas.0611554104.

Tronel, S., & Alberini, C. M. (2007). Persistent disruption of a traumatic memory by post-retrieval inactivation of glucocorticoid receptors in the amygdala. *Biological Psychiatry, 62*(1), 33–39. http://dx.doi.org/10.1016/j.biopsych.2006.09.009.

Tronel, S., Milekic, M. H., & Alberini, C. M. (2005). Linking new information to a reactivated memory requires consolidation and not reconsolidation mechanisms. *PLoS Biology, 3*(9). http://dx.doi.org/10.1371/journal.pbio.0030293. e293.

Tronson, N. C., Wiseman, S. L., Olausson, P., & Taylor, J. R. (2006). Bidirectional behavioral plasticity of memory reconsolidation depends on amygdalar protein kinase A. *Nature Neuroscience, 9*(2), 167–169. http://dx.doi.org/10.1038/nn1628.

Valjent, E., Corbillé, A.-G., Bertran-Gonzalez, J., Hervé, D., & Girault, J.-A. (2006). Inhibition of ERK pathway or protein synthesis during reexposure to drugs of abuse erases previously learned place preference. *Proceedings of the National Academy of Sciences of the United States of America, 103*(8), 2932–2937. http://dx.doi.org/10.1073/pnas.0511030103.

Vetere, G., Restivo, L., Cole, C. J., Ross, P. J., Ammassari-Teule, M., Josselyn, S. A., & Frankland, P. W. (2011). Spine growth in the anterior cingulate cortex is necessary for the consolidation of contextual fear memory. *Proceedings of the National Academy of Sciences of the United States of America, 108*(20), 8456–8460, 1016275108 10.1073/pnas.1016275108.

von Hertzen, L. S., & Giese, K. P. (2005). Memory reconsolidation engages only a subset of immediate-early genes induced during consolidation. *Journal of Neuroscience, 25*(8), 1935–1942, 25/8/1935 10.1523/JNEUROSCI.4707-04.2005.

Wiltgen, B. J., Brown, R. A., Talton, L. E., & Silva, A. J. (2004). New circuits for old memories: the role of the neocortex in consolidation. *Neuron, 44*(1), 101–108. http://dx.doi.org/10.1016/j.neuron.2004.09.015 S0896627304005793.

Wiltgen, B. J., & Silva, A. J. (2007). Memory for context becomes less specific with time. *Learning & Memory, 14*(4), 313–317. http://dx.doi.org/10.1101/lm.430907.

Winocur, G., Moscovitch, M., & Bontempi, B. (2010). Memory formation and long-term retention in humans and animals: convergence towards a transformation account of hippocampal-neocortical interactions. *Neuropsychologia, 48*(8), 2339–2356, S0028-3932(10)00162-4 10.1016/j.neuropsychologia.2010.04.016.

Winocur, G., Moscovitch, M., & Sekeres, M. (2007). Memory consolidation or transformation: context manipulation and hippocampal representations of memory. *Nature Neuroscience, 10*(5), 555–557, nn1880 10.1038/nn1880.

Winters, B. D., Tucci, M. C., & DaCosta-Furtado, M. (2009). Older and stronger object memories are selectively destabilized by reactivation in the presence of new information. *Learning & Memory, 16*(9), 545−553. http://dx.doi.org/10.1101/lm.1509909.

Xue, Y.-X., Luo, Y.-X., Wu, P., Shi, H.-S., Xue, L.-F., Chen, C., & Lu, L. (2012). A memory retrieval-extinction procedure to prevent drug craving and relapse. *Science, 336*(6078), 241−245. http://dx.doi.org/10.1126/science.1215070.

Yehuda, R., Tischler, L., Golier, J. A., Grossman, R., Brand, S. R., Kaufman, S., & Harvey, P. D. (2006). Longitudinal assessment of cognitive performance in Holocaust survivors with and without PTSD. *Biological Psychiatry, 60*(7), 714−721. http://dx.doi.org/10.1016/j.biopsych.2006.03.069.

Zhang, Y., Fukushima, H., & Kida, S. (2011). Induction and requirement of gene expression in the anterior cingulate cortex and medial prefrontal cortex for the consolidation of inhibitory avoidance memory. *Molecular Brain, 4*, 4, 1756-6606-4-4 10.1186/1756-6606-4-4.